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JAMB Economics: The Complete Course

Concepts, worked examples and exercises for the UTME

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Chapter 1: Basic Concepts of Economics

Every morning, millions of Nigerians make economic decisions without calling them by that name. A trader in Onitsha Main Market decides how many cartons of provisions she can afford to restock. A student in Ibadan decides whether to spend his remaining pocket money on data or on a meal. A state government decides whether its limited revenue should build a road or equip a hospital. In each case, people want more than they can have, and so they must choose. Economics is the systematic study of that predicament and of the ways in which individuals, firms and governments respond to it.

This opening chapter lays the foundation on which the rest of the course is built. It explains what economics is, why scarcity forces choice, how choice is measured through opportunity cost, how the production possibility curve pictures the limits of an economy, what the factors of production are, and how different societies organise themselves to answer the basic economic questions. The vocabulary introduced here appears again and again in later chapters and in the UTME, so it must be learnt with care.

1.1 Definition and scope of economics

Several definitions of economics have been offered over the last two centuries, and each emphasises a different aspect of the subject. Early writers described it as the study of wealth, meaning the production, distribution and consumption of goods. Alfred Marshall later described it as the study of mankind in the ordinary business of life, with emphasis on material welfare. The most widely accepted modern definition, associated with Lionel Robbins, states that **economics is the science which studies human behaviour as a relationship between ends and scarce means which have alternative uses**.

Each part of Robbins' definition deserves attention. The *ends* are human wants, which are unlimited. The *means* are the resources available for satisfying those wants, and these are limited. The phrase *alternative uses* reminds us that the same resource can be employed in different ways: a hectare of land in Benue may grow yams, or soya beans, or hold a warehouse. Because wants exceed resources and resources can be put to different uses, a society cannot avoid choosing, and the study of that choosing is economics.

> **Key point.** Economics is a social science because it studies people in society, and a science because it uses a method of observation, hypothesis and testing. It is not simply the study of money. Money is only one of many things that economists study.

Scope: microeconomics and macroeconomics

Microeconomics studies the behaviour of individual decision-makers and the markets in which they meet. The price of a bag of cement, the output of a single bakery and the demand of one household for petrol are microeconomic topics. **Macroeconomics** studies the economy as a whole, dealing with aggregates such as national income, the general price level, total employment, the money supply and the balance of payments. A fall in the price of tomatoes in Mile 12 Market is a micro matter; a rise in the national inflation rate from 15% to 22% is a macro matter.

Positive and normative statements

A **positive statement** describes what is, was or will be, and it can in principle be tested against facts. “An increase in the price of rice reduces the quantity of rice demanded” is positive. A **normative statement** expresses a value judgement about what ought to be, and cannot be settled by facts alone. “The government ought to subsidise rice farmers” is normative, because it depends on what one thinks is fair or desirable. The distinction matters in the UTME, which often asks candidates to identify which of several statements is positive. A useful test is to ask whether the statement could be shown to be true or false by evidence. If so, it is positive; if it contains words such as *should*, *ought*, *better* or *fair*, it is probably normative.

1.2 Human wants and the problem of scarcity

A **want** is a desire for a good or service. It is a desire that a person would like to satisfy if resources allowed, and it may or may not be backed by the ability to pay. Human wants have several important characteristics. They are *unlimited*, because as soon as one want is satisfied another arises. They are *recurring*, since hunger returns after each meal. They are *competing*, because income spent on one thing cannot be spent on another. And they are *satiable individually* but not collectively: a person may be completely satisfied with the fourth plate of rice, yet he will soon want something else.

Wants are commonly classified as follows. **Necessities** are those wants that must be met for survival, such as food, clothing and shelter. **Comforts** raise the quality of life above the bare minimum, such as a fan or a mattress. **Luxuries** are costly wants not required for comfort, such as a sports car or designer jewellery. The classification is not fixed: a mobile phone was once a luxury and is now, for many traders and students, close to a necessity.

Scarcity

Scarcity is the fundamental economic problem. It means that resources are insufficient to satisfy all human wants. It is important to understand that scarcity is a relative concept: it does not mean that something is rare. Air is plentiful but not scarce in the economic sense, because it is free and everyone can have as much as they wish. Sand may be abundant in a quarry, yet it is economically scarce if it takes effort and

money to obtain it and people compete for it. A good is scarce when the amount available at a price of zero is less than the amount people would like to have.

> **Exam tip.** Do not confuse scarcity with shortage. Scarcity is a permanent condition affecting every society, rich or poor. A shortage is a temporary situation in a particular market where quantity demanded exceeds quantity supplied at the going price, for example a shortage of petrol before a fuel-price announcement.

Goods and services

The things that satisfy wants are called **goods** when they are physical and **services** when they are intangible acts, such as teaching, transportation or banking. Goods may be **free goods**, which exist in such abundance that no price is charged (sunshine, ocean air), or **economic goods**, which are scarce and command a price. They may be **consumer goods**, bought for final use (bread, shoes), or **capital goods**, used to produce other goods (tractors, grinding machines). Consumer goods are further divided into **durable** goods, which last for a long time (a refrigerator) and **non-durable** goods, which are used up quickly (bread, bottled water). Goods may also be **public goods**, whose benefits are enjoyed by all whether or not they pay (street lighting, national defence), or **private goods**, which can be owned and enjoyed by one person to the exclusion of others.

1.3 Choice, scale of preference and opportunity cost

Since resources cannot satisfy every want, people must **choose**. Rational choice requires an order of priority. A **scale of preference** is a list of a person's wants arranged in descending order of importance. Suppose Amina has ₦20,000 and her wants, in order, are: school fees balance, a textbook set, a pair of shoes, a new phone case. She satisfies the first with her limited money, then the next, and so on until the money is exhausted. The wants that remain unsatisfied are the ones lowest on her scale.

A scale of preference differs from person to person and changes over time with age, income, culture and advertising. A farmer in Kebbi and a banker in Victoria Island will rank wants differently. A scale of preference applies equally to a household, a firm or a government, though the items ranked are different.

Opportunity cost

Whenever one want is satisfied at the expense of another, something is given up. The **opportunity cost** of a decision is the value of the next best alternative that is sacrificed. It is a concept of real cost rather than money cost. If Chidi spends Saturday afternoon working at his uncle's shop, where he earns ₦3,000, rather than attending a paid tutorial that would have earned him ₦4,000, his opportunity cost is the ₦4,000 he did not earn, not simply the effort of working.

Key point. Opportunity cost is the *next best* alternative forgone, not the sum of all alternatives forgone. If a person chooses among three options, the opportunity cost of choosing one is the value of the best of the other two.

WORKED EXAMPLE 1.1

Mrs Bello has ₦50,000 and can use it in only one of four ways: (A) buy a sewing machine, expected to bring a net gain of ₦18,000 per year; (B) deposit it in a savings account yielding 12% interest per year; (C) pay a part of her son's fees, from which no money return arises; (D) buy stock for her stall expected to give a net gain of ₦15,000 per year. If she chooses the sewing machine, what is her opportunity cost?

Solution: Compute the yearly return of each alternative. Option B gives 12% of ₦50,000 = 0.12 × 50,000 = ₦6,000. Option D gives ₦15,000. Option C gives no money return, although it has value of another kind. Among the alternatives to the sewing machine, the best money return is option D at ₦15,000, and this is greater than B. Therefore the opportunity cost of choosing the sewing machine is the ₦15,000 yearly return from stock, in money terms (the benefit of paying fees is non-monetary and is not ranked here). The three forgone options are not added together.

1.4 The production possibility curve

An economy’s resources are limited, and so at any time there is a limit on how much it can produce. The **production possibility curve** (PPC), also called the production possibility frontier or the transformation curve, shows the different combinations of two goods that an economy can produce when all its resources are fully and efficiently employed and technology is fixed. It is a tool for illustrating scarcity, choice and opportunity cost in a single picture.

Consider an economy producing only maize and cement. The table below shows the maximum quantity of cement (in thousand tonnes) that can be made for each quantity of maize (in thousand tonnes).

Combination	Maize (thousand tonnes)	Cement (thousand tonnes)
A	0	100
B	20	96
C	40	84
D	60	64
E	80	36
F	100	0

Plotting these points with maize on the horizontal axis and cement on the vertical axis gives a curve that bends outward from the origin, called concave to the origin. Several lessons arise from the picture.

- **Points on the curve** represent efficient, full employment of resources. The economy is making the most it can of the two goods.
- **Points inside the curve** represent unemployment of resources or inefficient use. The economy could produce more of one good without giving up any of the other.
- **Points outside the curve** are unattainable with present resources and technology.
- Moving along the curve from one point to another shows the **opportunity cost**: to gain more of one good, some of the other must be given up.

Why the curve is concave

The curve bends because of the **law of increasing opportunity cost**. Resources are not equally suited to every use. Land best suited to farming will make poor factory sites, and workers skilled at bricklaying are not equally good at harvesting. When the economy first shifts from cement to maize, it moves the resources that are best at growing maize, so little cement is lost for each tonne of maize gained. As the shift continues, resources that are increasingly ill-suited to maize growing must be transferred, so ever more cement is lost for each extra tonne of maize.

WORKED EXAMPLE 1.2

Using the table above, calculate the opportunity cost of producing 20 additional thousand tonnes of maize when the economy moves (i) from A to B, (ii) from C to D and (iii) from E to F. Express each as the cement lost per tonne of maize gained.

Solution: (i) From A to B, cement falls from 100 to 96, a loss of 4 thousand tonnes, for a gain of 20 thousand tonnes of maize. Opportunity cost per tonne of maize = 4/20 = 0.2 tonne of cement. (ii) From C to D, cement falls from 84 to 64, a loss of 20, so the cost per tonne of maize is 20/20 = 1.0 tonne of cement. (iii) From E to F, cement falls from 36 to 0, a loss of 36, so the cost is 36/20 = 1.8 tonnes of cement. The cost per tonne of maize rises from 0.2 to 1.0 to 1.8, confirming increasing opportunity cost, which is why the curve is concave to the origin.

Shifts of the curve: economic growth

The curve can shift. An increase in the quantity or quality of resources, or an improvement in technology, shifts the entire curve outward to the right and represents **economic growth**. Discovery of new oil fields, a more educated labour force, improved seed varieties and the building of new roads all push the frontier outward. A destruction of resources by war, flood or disease would shift it inward. If the improvement affects only one good, say a new high-yield maize seed, the curve swings outward on the maize axis only, with the intercept on the cement axis unchanged.

Exam tip. A straight-line PPC means constant opportunity cost, which happens only when resources are perfectly adaptable between the two uses. The usual curve is concave, and a curve convex to the origin would imply decreasing opportunity cost, which is unusual and not normally expected.

1.5 Factors of production

Goods and services are produced by combining resources called **factors of production**. Economists recognise four.

Land

In economics, **land** means all natural resources provided free by nature, not merely the surface of the earth. It includes soil, forests, rivers, minerals, crude oil, the sea and its fish, and climate. Its reward is **rent**. Land has distinctive characteristics: its total supply is fixed in the aggregate (it is not produced by effort), it is immobile geographically, and it differs in quality and fertility from one place to another. Nigeria's deposits of crude oil in the Niger Delta, tin on the Jos Plateau and limestone in Ewekoro are examples of land in this sense.

Labour

Labour is any physical or mental human effort used in production in return for a reward. The reward is **wages or salaries**. A teacher, a bricklayer, a doctor and a tailor all supply labour. Labour is distinctive

because it cannot be separated from the worker, it is perishable (a day of unused labour is lost forever), it varies in skill, and its supply is influenced by population, education and training. The efficiency of labour can be raised by training and by the division of labour, a topic treated in the chapter on production.

Capital

Capital is man-made wealth that is used to produce further goods and services: machinery, buildings, tools, roads and vehicles used for business. Its reward is **interest**. Capital is created by saving and investing, that is, by giving up present consumption to build productive equipment for the future. Capital may be *fixed* (lasting for many production cycles, like a factory) or *circulating* (used up in one production cycle, like flour in a bakery). Money itself is not capital in the strict sense, because it does not directly produce anything; it is the means of buying capital goods.

Entrepreneurship

The **entrepreneur** is the organiser who brings together land, labour and capital, takes the decisions about what to produce and how, and bears the risk of loss. The reward is **profit**. A person who opens a block-making business in Abeokuta, hires workers, buys a moulding machine and risks his own savings is performing the entrepreneurial function. Entrepreneurs make four main contributions: organisation, decision-making, innovation and risk-bearing.

Factor	Meaning	Reward	Example
Land	Natural resources	Rent	Farmland, crude oil
Labour	Human effort	Wages and salaries	Nurse, welder
Capital	Man-made aids to production	Interest	Bakery oven
Entrepreneur	Organisation and risk-taking	Profit	Business owner

Mobility of factors

Factors differ in how easily they can be moved. *Geographical mobility* is the ease of moving from place to place; *occupational mobility* is the ease of changing from one use to another. Labour is more mobile than land, but even labour finds movement difficult because of family ties, language, cost of relocation and the need for new skills. The lower the mobility, the greater the cost of shifting resources between uses, and this partly explains why the production possibility curve bends.

1.6 The basic economic problems

Because resources are scarce, every society, whatever its wealth or political system, must answer the same set of questions.

1. **What to produce, and in what quantities?** Society must decide how much of its resources should go to food, how much to housing, to defence, to education, and so on. This is the problem of the

allocation of resources, and it also involves choosing between consumer goods now and capital goods for growth later.

2. **How to produce?** There are different techniques. Cassava may be harvested by many workers with hoes (labour-intensive) or by few workers with machines (capital-intensive). The choice depends on the relative prices and availability of factors.
3. **For whom to produce?** Society must decide how output is shared among its members: by income, need, social rank or some other rule. This is the problem of *distribution*.
4. **How to sustain and grow the economy?** A society must decide how much of its current output to save and invest so that future output rises, and must also ensure that its resources are fully employed. Some writers present these as the problems of economic growth and of full employment.

1.7 Economic systems

An **economic system** is the set of institutions and arrangements by which a society answers the basic questions. Four main types are recognised.

Traditional economy

In a traditional economy, decisions are made according to custom, inheritance and long-established ways. Sons farm as their fathers did; tools are simple; production is largely for subsistence. Some rural

communities in Nigeria still show traditional features. It is stable and rarely wasteful in resource use, yet it is slow to adopt new methods, and living standards stay low.

Market (capitalist or free enterprise) economy

In a market economy, resources are privately owned, and decisions are made by individuals and firms responding to prices determined through the interaction of buyers and sellers. The profit motive guides producers, and consumer spending acts as “votes” telling firms what to make. Features include private property, freedom of enterprise, freedom of choice, competition and a limited role for the state. Strengths are efficiency, innovation and wide consumer choice. Weaknesses include inequality of income, neglect of public goods, the tendency towards monopoly, and instability through booms and slumps. No economy is purely of this type; the United States approximates it.

Command (planned or socialist) economy

In a command economy, the means of production are owned mainly by the state, and a central planning authority decides what, how and for whom to produce. The old Soviet Union and present-day Cuba and North Korea have used this method. Its strengths are the ability to concentrate resources on national goals, a more equal sharing of income, and the avoidance of wasteful duplication. Its weaknesses include lack of incentive, bureaucratic delays, shortages and surpluses caused by planning errors, little consumer choice, and limited freedom.

Mixed economy

Most countries, including Nigeria, Ghana, India and the United Kingdom, operate a **mixed economy**, in which private enterprise and government both play important roles. The private sector produces most consumer goods, while the government provides public goods, regulates industry, runs some key enterprises, and redistributes income through taxes and social spending. The balance between the two varies widely from country to country and from period to period. In Nigeria, privatisation and commercialisation programmes since the 1980s have shifted the balance toward the private sector, yet the government still dominates crude oil revenue and provides most basic education and infrastructure.

Feature	Market	Command	Mixed
Ownership	Mainly private	Mainly state	Both
Decision-maker	Price mechanism	Central planners	Prices plus government
Motive	Profit	National plan	Profit and social welfare
Main weakness	Inequality	Inefficiency	Conflict of objectives

1.8 Economic methodology and tools

Economists use a few simple tools repeatedly, and the UTME tests candidates on their use.

Economic models and ceteris paribus

An **economic model** is a simplified representation of reality that isolates the important relationships. Because many things change at once in the real world, economists analyse one relationship at a time by assuming that all other relevant factors remain unchanged. This assumption is expressed by the Latin phrase ***ceteris paribus***, meaning “other things being equal”. When we say that a fall in price raises quantity demanded, we mean this holds when income, tastes and other prices do not change.

Tables, graphs and equations

Relationships can be set out as a table of values, drawn as a graph, or written as an equation. A **variable** is a quantity that can take different values; the **independent variable** influences the **dependent variable**. Economists conventionally plot price on the vertical axis even when it is the cause. The slope of a straight line equals the change in the vertical variable divided by the change in the horizontal variable.

Index numbers and percentage change

Percentage change = (new value − old value)/old value × 100. This is the basic calculation behind growth rates, inflation and elasticity. For example, if the price of garri rises from ₦400 to ₦500 per measure, the percentage change is (500 − 400)/400 × 100 = 100/400 × 100 = 25%.

WORKED EXAMPLE 1.3

A country's output of goods was 8,000 units in 2022 and 8,600 units in 2023. Its population rose from 2,000 to 2,050 over the same period. Find (i) the percentage growth in total output and (ii) the percentage change in output per head.

Solution: (i) Growth in output = (8,600 − 8,000)/8,000 × 100 = 600/8,000 × 100 = 7.5%. (ii) Output per head in 2022 = 8,000/2,000 = 4.0 units; in 2023 = 8,600/2,050 = 4.195 units (to 3 decimal places). Percentage change = (4.195 − 4.0)/4.0 × 100 = 4.88% approximately. Output per head grew more slowly than total output because population also grew.

Exercise 1.1

1. Define economics according to Lionel Robbins and explain the term "alternative uses".
2. Which of the following is a normative statement? (a) Inflation in Nigeria rose last year. (b) The government should reduce taxes on small firms. (c) A rise in the price of fuel increases transport fares. (d) Unemployment among graduates is high.
3. Distinguish between scarcity and shortage, giving one Nigerian example of each.
4. Emeka has ₦10,000 and can either buy a pair of shoes or save the money for a year at 10% interest. Calculate the interest he gives up by buying the shoes.

5. Using the PPC table in section 1.4, state the opportunity cost in cement of increasing maize output from 60 to 80 thousand tonnes.
6. State two reasons why the production possibility curve is concave to the origin.
7. Identify the factor of production and the reward in each case: (a) a plumber in Enugu; (b) a fishing pond in Ondo; (c) a delivery van owned by a bakery; (d) a woman who starts a catering business and hires four staff.
8. A country's output rises from 5,000 to 5,600 units while its population rises from 1,000 to 1,050. Calculate the percentage growth in output per head, to one decimal place.
9. Explain three differences between a market economy and a command economy.
10. Why is Nigeria described as a mixed economy? Give two features of each sector.

Answers to Exercise 1.1

1. Economics is the science that studies human behaviour as a relationship between unlimited ends and scarce means which have alternative uses; alternative uses means that a given resource, such as land, can be employed to produce different goods, so using it for one purpose means not using it for another.

2. (b). It contains “should” and expresses a value judgement; the others can be tested against facts.
3. Scarcity is the permanent condition that resources are insufficient for all wants (for example, limited places in federal universities); a shortage is a temporary excess of quantity demanded over quantity supplied at the going price (for example, petrol shortage during a supply disruption).
4. Interest forgone = 10% of 10,000 = ₦1,000.
5. At 60 maize, cement = 64; at 80 maize, cement = 36. Cement lost = 64 − 36 = 28 thousand tonnes.
6. Resources are not equally suited to producing both goods, so opportunity cost rises as more of one good is made; and factors are imperfectly mobile between uses.
7. (a) Labour, wages; (b) land, rent; (c) capital, interest; (d) entrepreneur (also hires labour), profit.
8. Output per head: 5,000/1,000 = 5.0; 5,600/1,050 = 5.333. Growth = (5.333 − 5.0)/5.0 × 100 = 6.7%.
9. Any three of: private versus state ownership; price mechanism versus central planning; profit motive versus plan targets; wide consumer choice versus limited choice; greater inequality versus more equal distribution; competition versus state monopoly.
10. Private enterprise and government both operate. Private sector: private ownership of firms, profit motive, price-guided markets. Public sector: government provision of roads and defence, state-owned enterprises and regulation.

Chapter summary

- Economics studies how people choose among alternatives when wants are unlimited and resources are scarce and have alternative uses.
- Microeconomics studies individual units and markets; macroeconomics studies aggregates such as national income and inflation. Positive statements can be tested; normative statements express value judgements.
- Scarcity is a permanent condition, different from a temporary shortage. Wants are unlimited, recurring and competing.
- Choice requires a scale of preference. Opportunity cost is the value of the next best alternative forgone.
- The production possibility curve shows the maximum combinations of two goods attainable with full employment and fixed technology; it is concave because of increasing opportunity cost, and it shifts outward with economic growth.
- The factors of production are land (rent), labour (wages), capital (interest) and entrepreneurship (profit).
- Every society must decide what, how and for whom to produce, and how to grow. Traditional, market, command and mixed systems answer these questions differently; Nigeria is a mixed economy.

- Ceteris paribus, percentage change and simple graphs are the basic tools of economic analysis.

Chapter 2: Demand and Supply

Walk through Mile 12 market in Lagos on a December morning and you will see the central idea of this chapter at work. Tomatoes that sold for a modest sum in the rainy season now cost far more, because transport from the north is difficult and the harvest is smaller; at the same time, festive shoppers are crowding the stalls. Sellers raise their prices, some buyers grumble and buy fewer baskets, and a few traders drive to farther towns to bring in more stock. Nobody in the market is issuing instructions, yet quantities and prices adjust. The tools economists use to describe this adjustment are **demand** and **supply**.

Demand and supply analysis is the single most important technique in microeconomics, and it appears in nearly every UTME paper in one form or another: a schedule to read, a curve to interpret, a shift to identify, an equilibrium to calculate. This chapter builds the theory carefully, beginning with the buyer, then the seller, and then bringing the two together to explain how a market price is determined and how it changes. We close with the effects of government price control and the idea of surplus.

2.1 The meaning of demand

In everyday speech, to "demand" something means to insist upon it. In economics the word has a precise meaning. **Demand** is the quantity of a good or service that consumers are willing and able to buy at each

possible price during a given period of time, other things remaining equal. Three parts of this definition deserve emphasis.

First, demand requires *willingness and ability* to pay. A secondary-school pupil in Onitsha may greatly desire a new laptop, but if she has no money she exerts no effective demand for it. Economists call a want backed by purchasing power **effective demand**. A want without purchasing power is only a wish.

Second, demand is always related to a *price* and a *period*. To say "Nigerians demand 40 million bags of rice" tells us little; we must say how many bags per year at what price. Demand is therefore a relationship, not a single number.

Third, the phrase *other things remaining equal* (ceteris paribus) isolates the effect of price from the effects of all other influences.

The law of demand

The **law of demand** states that, other things being equal, the higher the price of a good, the smaller the quantity demanded, and the lower the price, the greater the quantity demanded. Price and quantity demanded move in opposite directions. Three reasons are usually given:

- **The income effect.** When the price of a good falls, the consumer's money income can buy more of it; real purchasing power has risen, so more is bought.
- **The substitution effect.** When the price of a good falls, it becomes cheaper relative to its substitutes, so consumers switch towards it. If

the price of garri falls while the price of rice stays the same, some households replace rice with garri.

- **Diminishing marginal utility.** Each additional unit of a good gives less extra satisfaction than the one before it. A consumer will buy additional units only if the price is lower.

There are some exceptions to the law. A **Giffen good** is an inferior good, usually a staple food, on which very poor consumers spend most of their income; when its price rises they may be forced to buy more of it because they can no longer afford better foods. A **snob or prestige good** (a rare jewel, an expensive car) may be bought more at a high price because the high price confers status. Speculative buying, when buyers expect further price rises, can also produce a rise in demand as the price rises. These cases are rare, and examiners treat them as exceptions to a general rule.

> **Key point.** A change in the *price of the good itself* causes a movement *along* the demand curve (an extension or contraction of demand). A change in any *other* determinant shifts the whole curve (an increase or decrease in demand). This distinction is tested almost every year.

2.2 The demand schedule and demand curve

A **demand schedule** is a table listing the quantities of a good demanded at various prices. A **demand curve** is the graph of that schedule, with price on the vertical axis and quantity on the horizontal axis. An **individual demand schedule** shows one consumer's behaviour; the **market demand schedule** is obtained by adding the quantities demanded by all consumers at each price (horizontal summation).

Price of a loaf (₦)	Adaasi (loaves per week)	Bayo (loaves per week)	Market (loaves per week)
800	1	0	1
700	2	1	3
600	3	2	5
500	4	3	7
400	5	4	9

The demand curve plotted from such a schedule slopes downward from left to right. Its slope is negative because of the law of demand. A straight-line demand curve can be written as an equation, for example $Q_d = 100 - 2P$, where Q_d is the quantity demanded and P the price. At $P = 10$ we have $Q_d = 80$; at $P = 30$, $Q_d = 40$. The negative coefficient of P expresses the inverse relationship.

2.3 Determinants of demand

The following factors, other than the good's own price, determine how much of a good consumers will buy. A change in any of them shifts the demand curve to the right (an **increase** in demand) or to the left (a **decrease** in demand).

1. **Consumers' income.** For a **normal good**, an increase in income raises demand. Examples are clothing, meat and electronic gadgets. For an **inferior good**, such as second-hand clothing sold in a "bend-down-select" market or a cheap brand of noodles, a rise in income reduces demand as consumers switch to better alternatives.
2. **Prices of related goods. Substitutes** are goods that satisfy the same want (Peak milk and Cowbell milk; tea and coffee; bus and train travel). If the price of one substitute rises, demand for the other rises. **Complements** are goods used together (bread and butter, cars and petrol, a generator and fuel). If the price of petrol rises, the demand for cars falls because the combined cost of ownership has increased.
3. **Tastes, fashion and habits.** A successful advertising campaign, a new fashion in Ankara designs or a health scare can shift demand sharply.
4. **Population size and structure.** A growing population raises market demand; a population with many young people demands more school books and toys, while an ageing population demands more medical services.
5. **Distribution of income.** If income is shifted from rich to poor, demand for basic goods rises while demand for luxury goods falls.

6. **Expectations.** If consumers expect the price of petrol to rise next week, they buy more today, so current demand increases.
7. **Seasons and climate.** Umbrellas and raincoats are demanded in the rainy season; fans and cold drinks in the harmattan-free hot months.
8. **Credit availability and interest rates.** Easy credit raises demand for durable goods such as vehicles and houses.
9. **Government policy.** Taxes on a product, subsidies for consumers, or a ban on certain goods affect demand.

> **Exam tip.** When a question says "an increase in the price of a substitute," draw the demand curve of the *other* good shifting right. When it says "an increase in the price of a complement," the demand for the *other* good shifts left. Be careful not to confuse the good whose price changed (which sees a movement along its own curve) with its partner (which sees a shift).

Types of demand

Economists distinguish several kinds of demand. **Joint demand** arises for complements, such as a vehicle and its tyres. **Composite demand** arises when a good has several uses: palm oil is demanded for cooking, soap making and industry, so the total demand is the sum of the demands for each use. **Derived demand** arises when a good is wanted for what it helps to produce; the demand for bricklayers is derived from the demand for buildings. **Competitive demand** exists between substitutes.

2.4 The meaning of supply

Supply is the quantity of a good or service that producers are willing and able to offer for sale at each possible price during a given period, other things being equal. As with demand, supply is a relationship between price and quantity. The **law of supply** states that, other things being equal, the higher the price of a good, the larger the quantity supplied, and the lower the price, the smaller the quantity supplied. Price and quantity supplied move in the same direction.

The reason is the profit motive. A higher price makes production more profitable, attracting existing firms to expand output and new firms to enter the industry. Also, because costs per unit generally rise as output expands (the law of diminishing returns, studied in Chapter 4), producers will only increase output if the price rises enough to cover the higher cost of the extra units.

A **supply schedule** lists quantities supplied at different prices. The **supply curve** slopes upward from left to right. A linear supply function might be written Qs = 20 + 3P: at P = 10, Qs = 50; at P = 20, Qs = 80. The market supply is the horizontal sum of the supplies of all firms.

Exceptions to the law of supply

Some goods do not follow the usual rule. The supply of **agricultural products** in the short run is fixed by the harvest, so supply cannot respond to price (the supply curve is vertical). The supply of **labour**

may bend backward at very high wages: when pay is already high, workers may choose more leisure. The supply of **rare works of art**, a famous painter's originals for instance, is fixed whatever the price.

2.5 Determinants of supply

A change in the price of the good itself causes a movement along the supply curve: an **extension** when the price rises and a **contraction** when it falls. A change in any other determinant shifts the whole curve: to the right for an **increase** in supply and to the left for a **decrease** in supply.

1. **Cost of production.** Higher wages, dearer raw materials, rising fuel prices or higher rent reduce profit at every price, so supply falls. Cheaper inputs raise supply.
2. **State of technology.** Improved methods lower the cost of production and raise supply. The introduction of mechanised rice mills in Kebbi increased the supply of milled rice.
3. **Prices of other goods.** A farmer may switch from growing maize to growing cassava if the price of cassava rises; the supply of maize falls. This is *competitive supply*. *Joint supply* arises when two goods are produced together, as with beef and hides: a rise in the price of beef increases the supply of hides.
4. **Number of firms.** Entry of new firms increases market supply.
5. **Taxes and subsidies.** A tax on producers raises costs and reduces supply; a subsidy, such as a fertiliser subsidy to farmers, lowers costs and increases supply.

6. **Weather and natural conditions.** Good rains increase the supply of crops; drought, flood or pest attack reduces it.
7. **Producers' expectations.** If sellers expect prices to rise in future, they may withhold stock now, reducing current supply.
8. **Political stability and infrastructure.** Insecurity on highways or the poor state of roads reduces the quantity that reaches the market.
9. **Goals of the firm.** A firm aiming at sales maximisation will supply more than one aiming at profit.

WORKED EXAMPLE 2.1

The demand and supply schedules for bags of rice in a Kano market are shown below. Find the equilibrium price and quantity, and state what exists at ₦40,000 and at ₦60,000 per bag.

Price per bag (₦)	Quantity demanded	Quantity supplied
40,000	500	200
45,000	450	300
50,000	400	400
55,000	350	500
60,000	300	600

Solution: Equilibrium occurs where quantity demanded equals quantity supplied, which is at ₦50,000 with 400 bags. At ₦40,000, demand (500) exceeds supply (200), so there is an excess demand (shortage) of 500 − 200 = 300 bags and price tends to rise. At ₦60,000, supply (600) exceeds demand (300), so there is an excess supply (surplus) of 600 − 300 = 300 bags and price tends to fall.

2.6 Market equilibrium

A market is in **equilibrium** when the quantity demanded equals the quantity supplied at the prevailing price; there is no tendency for price to change. This price is called the **equilibrium price** (or market-clearing

price) and the corresponding quantity the **equilibrium quantity**. On a diagram, it is the point where the demand and supply curves intersect.

If the price is above the equilibrium, quantity supplied exceeds quantity demanded: there is a **surplus** or excess supply. Unsold stock builds up, sellers cut prices, quantity demanded rises (a movement down the demand curve) and quantity supplied falls (a movement down the supply curve) until the surplus disappears. If the price is below equilibrium, there is a **shortage** or excess demand; buyers compete for the limited supply and bid the price up until equilibrium is restored. This self-correcting process is sometimes described as the "invisible hand" of the market.

Calculating equilibrium from equations

When demand and supply are given as equations, we set Qd = Qs and solve for P, then substitute to find Q.

WORKED EXAMPLE 2.2

In a market for pure water sachets, Qd = 900 − 4P and Qs = 200 + 3P, where P is in naira per bag and Q in thousands of bags. Find the equilibrium price and quantity.

Solution: At equilibrium, 900 − 4P = 200 + 3P. Collecting terms, 900 − 200 = 3P + 4P, so 700 = 7P and P = ₦100. Then Qd = 900 − 4(100) = 900 − 400 = 500. Check with supply: Qs = 200 + 3(100) = 500. The equilibrium quantity is 500 thousand bags at ₦100 per bag.

Effects of shifts in demand and supply

When a curve shifts, a new equilibrium is established. The results follow a predictable pattern:

Change	Effect on equilibrium price	Effect on equilibrium quantity
Demand increases (shift right)	Rises	Rises
Demand decreases (shift left)	Falls	Falls
Supply increases (shift right)	Falls	Rises
Supply decreases (shift left)	Rises	Falls

Consider a poor yam harvest after a flood in Benue State. Supply shifts left; with demand unchanged, the price of yam rises and the quantity traded falls. Conversely, when a new brand of imported tomato paste enters the market, the supply of tomato paste shifts right, lowering its price and raising the quantity bought. If demand and supply both change at once, one of the two effects (price or quantity) will be definite and the other will depend on the relative size of the shifts: for example, if both demand and supply increase, equilibrium quantity certainly rises, but price may rise, fall or stay the same.

WORKED EXAMPLE 2.3

Using the market of Worked example 2.2, suppose a rise in household incomes changes demand to Qd = 1,020 − 4P while supply is unchanged. Find the new equilibrium, and the change in price and quantity.

Solution: Set 1,020 − 4P = 200 + 3P. Then 820 = 7P and P = 117.14 (approximately ₦117.14). Quantity: Qs = 200 + 3(117.14) = 200 + 351.43 = 551.43 thousand bags. Check on demand: 1,020 − 4(117.14) = 1,020 − 468.57 = 551.43. The price rose from ₦100 to about ₦117.14, an increase of about 17.1%, and quantity rose from 500 to about 551.4 thousand bags, an increase of about 10.3%.

2.7 Government intervention: price ceilings and price floors

Governments sometimes believe the market price is unfair and fix it by law. A **price ceiling** (maximum price) is a legal limit above which the price may not rise. To be effective it must be set *below* the equilibrium price. The usual aim is to protect consumers: examples include controls on the pump price of petrol, on rent and on the prices of essential foods. The consequence is excess demand: at the lower price, consumers want more than producers will supply. Shortages, queues, rationing, black

markets and "black-market" prices above the legal limit result. Petrol queues in Nigerian cities are a familiar example of how a ceiling can create a shortage when the controlled price is below the cost of supply.

A **price floor** (minimum price) is a legal limit below which the price may not fall. To be effective, it must be set *above* equilibrium. It is used to protect producers, for example a guaranteed minimum price for grains or cocoa, and to protect workers through a minimum wage. The consequence is excess supply: at the higher price, producers supply more than consumers buy. A surplus of produce remains unsold unless the government buys it up and stores it (buffer stock) or exports it; in the labour market the surplus appears as unemployment among those who would work at lower wages.

WORKED EXAMPLE 2.4

Using Qd = 900 − 4P and Qs = 200 + 3P (Worked example 2.2), the government fixes a maximum price of ₦80. Find the quantity demanded, the quantity supplied and the size of the shortage. If instead a minimum price of ₦130 is imposed, find the surplus.

Solution: At P = 80 the ceiling is below the equilibrium of ₦100, so it is effective. Qd = 900 − 320 = 580; Qs = 200 + 240 = 440. Shortage = 580 − 440 = 140 thousand bags. At P = 130 the floor is above equilibrium. Qd = 900 − 520 = 380; Qs = 200 + 390 = 590. Surplus = 590 − 380 = 210 thousand bags.

Key point. A price control that lies on the "wrong" side of equilibrium has no effect. A ceiling above the equilibrium price, or a floor below it, changes nothing because the market price is already within the legal limit.

2.8 Consumer surplus and producer surplus

Many consumers would be prepared to pay more than the market price. **Consumer surplus** is the difference between what a consumer is willing to pay for a good and what he or she actually pays. On a diagram it is the triangular area under the demand curve and above the price line. **Producer surplus** is the difference between the price a producer receives and the minimum price he would have accepted; it is the area above the supply curve and below the price line. The total of the two measures the benefit the market gives to society. A price ceiling or floor usually reduces this total.

For a linear demand curve, consumer surplus can be calculated as the area of a triangle: ½ × base × height, where the base is the equilibrium quantity and the height is the gap between the demand-curve intercept on the price axis and the equilibrium price. In Worked example 2.2, demand Qd = 900 − 4P has a price intercept at Q = 0, so P = 225. The consumer surplus is ½ × 500 × (225 − 100) = ½ × 500 × 125 = 31,250 (in thousands of naira, since quantity is in thousands). Examination questions usually give simple linear curves that meet the price axis at a positive value, so that the surplus is a plain triangle and the calculation

is direct. For instance, if a supply curve is Qs = 4P − 40, it meets the price axis at P = 10; with an equilibrium price of ₦30 and quantity of 80, producer surplus = ½ × 80 × (30 − 10) = ₦800.

2.9 Joint application: an extended example

Consider the market for diesel in Port Harcourt. Suppose that the government removes a subsidy. Producers' and importers' costs rise, so supply shifts left. If demand is unchanged, price rises and quantity falls. Transport operators, who use diesel as an input, face higher costs, so the supply of bus journeys also shifts left; fares rise. Households respond by walking, car-pooling or moving closer to work. This chain illustrates that markets are linked: a shift in one market feeds into others through costs and substitutes. A student who can trace such chains using demand-and-supply diagrams has mastered the heart of the topic.

Exercise 2.1

1. State the law of demand and give two reasons why the demand curve slopes downward.
2. Which of the following causes a shift in the demand curve for bread: (a) a fall in the price of bread, (b) a rise in the price of flour, (c) a rise in consumers' incomes, (d) a fall in the price of margarine, a complement? Explain which are shifts and which are movements.
3. If the price of tea rises, what happens to the demand for coffee, a substitute, and for sugar, a complement of tea?

4. Given Qd = 120 − 5P and Qs = 30 + 4P, find the equilibrium price and quantity.
5. A market has Qd = 300 − 2P and Qs = 60 + 4P. Find the equilibrium price and quantity, and the excess demand at P = 30.
6. Explain how a good harvest affects the price and quantity of tomatoes sold in the market.
7. Distinguish between an increase in supply and an extension of supply.
8. With Qd = 500 − 10P and Qs = 100 + 10P, the government imposes a minimum price of ₦30. Calculate the surplus.
9. Define consumer surplus. If the demand curve is Qd = 80 − 2P, the price is ₦20 and 40 units are bought, calculate the consumer surplus.
10. What is a Giffen good? Why is it an exception to the law of demand?
11. Explain why an effective price ceiling leads to a black market.

Answers to Exercise 2.1

1. Other things equal, quantity demanded falls as price rises. Reasons: income effect, substitution effect (also diminishing marginal utility).
2. (b) raises the cost of bread, so it shifts supply, not demand; (c) shifts demand right (bread is normal); (d) a fall in the price of a complement raises demand for bread, shifting right. (a) is a

movement along the demand curve (an extension of demand), not a shift.

3. Demand for coffee increases (shifts right); demand for sugar decreases (shifts left), because less tea is bought.
4. $120 - 5P = 30 + 4P$ gives $90 = 9P$, $P = 10$; $Q = 120 - 50 = 70$ (check $30 + 40 = 70$).
5. $300 - 2P = 60 + 4P$ gives $240 = 6P$, $P = 40$; $Q = 300 - 80 = 220$. At $P = 30$: $Q_d = 240$, $Q_s = 180$, excess demand = 60.
6. Supply shifts right; with demand unchanged, price falls and quantity traded rises.
7. An increase in supply is a rightward shift of the whole curve caused by non-price factors; an extension is a movement along the curve caused by a rise in the good's own price.
8. Equilibrium: $500 - 10P = 100 + 10P$, $P = 20$, so ₦30 is effective. $Q_d = 500 - 300 = 200$; $Q_s = 100 + 300 = 400$; surplus = 200.
9. Consumer surplus is the difference between what a buyer is willing to pay and what is paid. Intercept price: $Q_d = 0$ at $P = 40$. $CS = \frac{1}{2} \times 40 \times (40 - 20) = \frac{1}{2} \times 40 \times 20 = 400$ (check: Q_d at $P = 20$ is $80 - 40 = 40$).
10. An inferior staple on which poor households spend most of their income; when its price rises the income effect is so strong that they buy more of it, so the quantity demanded rises with price.
11. At the controlled price, demand exceeds supply; unsatisfied buyers are willing to pay more, and sellers can obtain a higher price illegally, so a parallel illegal market develops.

Exercise 2.2

1. A consumer's demand schedule is: ₦200 → 10 units, ₦180 → 14 units, ₦160 → 18 units, ₦140 → 22 units. Write the equation of the demand curve in the form Q = a − bP.
2. Which of the following would shift the supply curve of garri to the left? (a) A rise in the price of garri; (b) a rise in the cost of cassava; (c) a fall in the price of garri; (d) a subsidy on cassava.
3. The market demand is Qd = 400 − 2P and supply is Qs = 4P − 80. Calculate the equilibrium price and quantity.
4. Following a change in tastes, demand in Question 3 becomes Qd = 460 − 2P. Calculate the new equilibrium price and quantity.
5. Give two examples of complementary goods and two of substitute goods found in a Nigerian household.
6. What happens to the equilibrium price and quantity if both demand and supply increase by the same amount?
7. Why is the supply of fresh tomatoes in the short run said to be inelastic with respect to price? State its effect on the shape of the supply curve.
8. Explain composite demand with the example of electricity.
9. If the government fixes a maximum price for rice below equilibrium, state two consequences.

Answers to Exercise 2.2

1. Each ₦20 fall raises quantity by 4, so slope is −4/20 = −0.2 per naira, b = 0.2. At P = 200, Q = 10, so a = 10 + 0.2 × 200 = 50. Q = 50 − 0.2P (check at 140: 50 − 28 = 22).
2. (b); higher input cost reduces supply. (a) and (c) are movements; (d) increases supply.
3. 400 − 2P = 4P − 80 gives 480 = 6P, P = 80; Q = 400 − 160 = 240 (check 320 − 80 = 240).
4. 460 − 2P = 4P − 80 gives 540 = 6P, P = 90; Q = 460 − 180 = 280.
5. Complements: bread and butter; rice and stew; substitutes: tea and coffee; Indomie and spaghetti (any valid pairs).
6. Quantity rises; the price may remain unchanged if the two shifts are equal in size horizontally, otherwise it depends on relative size.
7. Output is fixed by the harvest and cannot be increased quickly when price rises; the supply curve is nearly or completely vertical.
8. Electricity is demanded by households, factories and offices; total demand is the sum of those demands.
9. Shortages and queues; black markets and rationing, possibly reduced output over time.

Chapter summary

- Demand is the quantity consumers are willing and able to buy at each price in a given period, other things equal. The law of

demand states that price and quantity demanded are inversely related.

- A change in the good's own price causes a movement along the demand (or supply) curve; a change in any other determinant shifts the curve.
- Determinants of demand include income, prices of substitutes and complements, tastes, population, expectations and policy.
- The law of supply states that price and quantity supplied are directly related. Determinants of supply include costs, technology, number of firms, taxes and subsidies, weather and expectations.
- Equilibrium is where Qd = Qs. A price above equilibrium creates a surplus, a price below it a shortage.
- Rightward shift of demand raises price and quantity; rightward shift of supply lowers price and raises quantity.
- An effective price ceiling (below equilibrium) causes shortages; an effective price floor (above equilibrium) causes surpluses.
- Consumer surplus is the gap between willingness to pay and the price paid; producer surplus is the gap between the price received and the minimum acceptable price.

Chapter 3: Elasticity of Demand and Supply

Suppose the price of a litre of petrol in Abuja rises by ten per cent. Will motorists buy a little less, or a great deal less? Suppose instead the price of salt rises by ten per cent. Will housewives notice? The law of demand tells us only that quantity demanded will fall; it says nothing about *how far*. Yet in practice the size of the response is what matters to a government deciding whether to tax a product, to a firm deciding whether to cut its price, and to a farmer wondering whether a bumper harvest will make him richer or poorer.

Elasticity is the measure of responsiveness. It tells us by what percentage one variable changes when another changes by one per cent. In this chapter we study four elasticities: price elasticity of demand, income elasticity of demand, cross elasticity of demand, and price elasticity of supply. We learn how to calculate each, how to interpret the result, what determines its size, and how businesses and governments use it. Elasticity calculations are a favourite of examiners, because they combine arithmetic with economic reasoning, so we work through many examples.

3.1 Price elasticity of demand

Price elasticity of demand (PED) measures the degree to which the quantity demanded of a good responds to a change in its price. It is defined as:

PED = percentage change in quantity demanded ÷ percentage change in price

Written with symbols, PED = (ΔQ/Q) ÷ (ΔP/P) = (ΔQ/ΔP) × (P/Q), where ΔQ is the change in quantity, ΔP the change in price, and Q and P the original quantity and price. Because the law of demand means that ΔQ and ΔP have opposite signs, PED is nearly always negative. Economists therefore usually ignore the minus sign and speak of the *absolute value* of elasticity. We shall do the same, unless a question asks otherwise.

WORKED EXAMPLE 3.1

When the price of a bag of cement in Ibadan rises from ₦8,000 to ₦8,800, the quantity demanded falls from 500 bags to 450 bags per week. Calculate the price elasticity of demand.

Solution: Percentage change in quantity = (450 − 500)/500 × 100 = −50/500 × 100 = −10%. Percentage change in price = (8,800 − 8,000)/8,000 × 100 = 800/8,000 × 100 = 10%. PED = −10 ÷ 10 = −1. The absolute value is 1, so demand is of unitary elasticity.

The arc (midpoint) formula

When the changes are large, the answer depends on whether we measure from the old or the new values. To avoid this, we may use the average (midpoint) of the two values as the base:

$$PED = [\Delta Q \div ((Q_1 + Q_2)/2)] \div [\Delta P \div ((P_1 + P_2)/2)]$$

UTME questions normally use the simple formula with the original values, so use the midpoint formula only when the question tells you to.

WORKED EXAMPLE 3.2

The price of a loaf of bread falls from ₦1,000 to ₦800 and quantity demanded rises from 300 to 420 loaves. Calculate PED by (a) the simple formula and (b) the midpoint formula.

Solution: (a) %ΔQ = (420 − 300)/300 × 100 = 40%. %ΔP = (800 − 1,000)/1,000 × 100 = −20%. PED = 40 ÷ (−20) = −2; absolute value 2. (b) Average quantity = (300 + 420)/2 = 360; ΔQ/average = 120/360 = 0.3333. Average price = (1,000 + 800)/2 = 900; ΔP/average = −200/900 = −0.2222. PED = 0.3333 ÷ (−0.2222) = −1.5; absolute value 1.5.

Notice that the two methods give different answers (2 and 1.5), though both show demand is elastic. This difference is why the method must be specified.

3.2 Interpreting the value of PED

The absolute value of PED places demand into one of five categories.

Value of |PED|	Name	Meaning	Shape of curve
0	Perfectly inelastic	Quantity does not change at all when price changes	Vertical line
Between 0 and 1	Inelastic	Percentage change in quantity is smaller than percentage change in price	Steep
Exactly 1	Unitary elastic	Percentage change in quantity equals percentage change in price	Rectangular hyperbola
Greater than 1	Elastic	Percentage change in quantity exceeds percentage change in price	Flat
Infinity	Perfectly elastic	Buyers will buy any quantity at one price but nothing at a higher price	Horizontal line

Perfectly inelastic demand is approximated by a life-saving drug for a patient, such as insulin for a diabetic. Perfectly elastic demand is approximated by the demand facing a single small farmer who sells yams in a market where the going price is fixed by thousands of others: if he raises his price even slightly, he sells nothing, and he has no reason to charge less.

Key point. A straight-line demand curve does *not* have the same elasticity all along its length. Elasticity is high at the top, where price is high and quantity is small, equals 1 at the midpoint, and is low at the bottom. Slope and elasticity are different ideas: slope measures absolute changes, elasticity measures percentage changes.

3.3 Determinants of price elasticity of demand

1. **Availability of substitutes.** The more and the closer the substitutes, the more elastic the demand. If the price of one brand of bottled water rises, buyers switch to another brand, so demand for each brand is elastic; demand for water in general is inelastic.
2. **Necessity or luxury.** Necessities such as salt, kerosene for the poor, and basic medicines have inelastic demand; luxuries such as foreign holidays and designer clothes have elastic demand.
3. **Proportion of income spent.** A good that takes a tiny share of income (matches, a pencil) is demanded inelastically because a price change is hardly noticed. A good that takes a large share (a car, rent) is elastic.
4. **Time period.** Demand is more elastic in the long run than in the short run because consumers need time to find substitutes or change habits. When petrol prices rise, motorists at first keep driving, but over several years they buy smaller cars or move closer to work.

5. **Habit and addiction.** Goods such as cigarettes have inelastic demand.
6. **Breadth of definition of the good.** "Food" is inelastic; "Golden Penny spaghetti" is elastic.
7. **Possibility of postponing purchase.** Purchases that can be delayed, such as buying new furniture, are more elastic than urgent ones such as a medical operation.

3.4 Elasticity and total revenue

A firm's **total revenue** (TR) is price multiplied by quantity sold: TR = P × Q. How total revenue responds to a price change depends on elasticity.

- If demand is **elastic**, a price *cut* raises total revenue, because the percentage rise in quantity is greater than the percentage fall in price. A price *rise* lowers total revenue.
- If demand is **inelastic**, a price *rise* raises total revenue, because quantity falls proportionally less than price rises. A price cut lowers total revenue.
- If demand is **unitary**, total revenue is unchanged by a price change.

WORKED EXAMPLE 3.3

A supermarket in Lekki sells 200 packets of biscuits per day at ₦500 each. It cuts the price to ₦400 and sales rise to 300 packets. Calculate PED and state the effect on total revenue. Would the supermarket be wise to continue cutting prices?

Solution: Original TR = 500 × 200 = ₦100,000. New TR = 400 × 300 = ₦120,000, so revenue rose by ₦20,000. %ΔQ = 100/200 × 100 = 50%. %ΔP = −100/500 × 100 = −20%. PED = 50 ÷ (−20) = −2.5, absolute value 2.5, which is elastic. Since demand is elastic, a price cut raises revenue, consistent with the figures. The supermarket should cut further only while demand remains elastic; as price falls along a straight-line demand curve, elasticity falls, and beyond the point where it reaches 1, further cuts reduce revenue (and costs must also be considered).

Applications

Understanding the relation between elasticity and revenue explains much business behaviour. Airlines offer cheap off-peak tickets to price-sensitive leisure travellers (elastic demand) while charging high fares to business travellers (inelastic demand). Governments impose heavy taxes on cigarettes and alcohol, whose demand is inelastic, to raise revenue; a tax on a product with elastic demand would cause sales to collapse and yield little. Farmers' incomes can fall after a bumper harvest: demand for staple foods is inelastic, so the large increase in supply causes a big

fall in price and little increase in quantity bought, so revenue falls. That is why governments sometimes introduce buffer stocks or minimum prices for agricultural produce.

3.5 Income elasticity of demand

Income elasticity of demand (YED) measures the responsiveness of the quantity demanded to a change in consumers' income:

YED = percentage change in quantity demanded ÷ percentage change in income

The sign matters. A **positive** YED means a normal good. If YED is greater than 1, the good is a **luxury** (income-elastic): quantity demanded rises faster than income, as with foreign travel or jewellery. If YED lies between 0 and 1, the good is a **necessity** (income-inelastic): demand rises less than income, as with salt and basic foods. A **negative** YED means an **inferior good**: demand falls as income rises, as with low-quality second-hand goods or very cheap grades of rice. A YED of zero means demand does not depend on income.

WORKED EXAMPLE 3.4

When Mrs. Okafor's monthly income rises from ₦200,000 to ₦250,000, her monthly purchases of chicken rise from 4 kg to 6 kg, and her purchases of cheap tinned fish fall from 10 tins to 8 tins. Calculate the income elasticity of demand for each good and classify them.

Solution: %ΔIncome = 50,000/200,000 × 100 = 25%. For chicken: %ΔQ = 2/4 × 100 = 50%; YED = 50 ÷ 25 = +2. Because YED is positive and above 1, chicken is a luxury (normal) good for her. For tinned fish: %ΔQ = −2/10 × 100 = −20%; YED = −20 ÷ 25 = −0.8. A negative YED means tinned fish is an inferior good.

YED is important for forecasting. As Nigeria's income rises, industries whose products have high YED, such as telecommunications, banking and mobile entertainment, can expect rapid growth, whereas agriculture of staple crops, with low YED, grows more slowly. This is why the share of agriculture in national output tends to fall as a country develops.

3.6 Cross elasticity of demand

Cross elasticity of demand (XED) measures the responsiveness of the quantity demanded of one good (A) to a change in the price of another good (B):

XED = percentage change in quantity demanded of A ÷ percentage change in price of B

If XED is **positive**, the goods are **substitutes**: a rise in the price of B drives buyers to A. The larger the value, the closer the substitutes. If XED is **negative**, the goods are **complements**: a rise in the price of B reduces the demand for both. If XED is **zero** or close to it, the goods are unrelated.

WORKED EXAMPLE 3.5

The price of a bottle of Brand B cola rises from ₦400 to ₦480, and the quantity of Brand A cola demanded rises from 1,000 to 1,300 bottles. Find the cross elasticity and state the relationship. Then find XED between petrol and cars if a 10% rise in petrol price reduces demand for cars by 4%.

Solution: %ΔP of B = 80/400 × 100 = 20%. %ΔQ of A = 300/1,000 × 100 = 30%. XED = 30 ÷ 20 = +1.5. Positive, so the colas are substitutes. For petrol and cars: XED = −4 ÷ 10 = −0.4. Negative, so they are complements, though the relationship is weak.

Firms use cross elasticity to understand how closely rivals' products compete. Two telecommunications companies whose services have a high positive XED will watch each other's tariffs closely; a firm producing complements will think about pricing them together, as printers and ink cartridges are sold.

3.7 Price elasticity of supply

Price elasticity of supply (PES) measures the responsiveness of quantity supplied to a change in price:

PES = percentage change in quantity supplied ÷ percentage change in price

Because the supply curve slopes upwards, PES is positive. If PES > 1, supply is elastic; if 0 < PES < 1, supply is inelastic; if PES = 1, supply is unitary; PES = 0 is perfectly inelastic (vertical supply curve) and PES = ∞ is perfectly elastic (horizontal supply curve).

A useful geometric rule: a *straight-line supply curve passing through the origin* has a PES of exactly 1 at every point. A straight supply curve that cuts the *price axis* (positive price intercept) is elastic (PES > 1); one that cuts the *quantity axis* is inelastic (PES < 1).

WORKED EXAMPLE 3.6

When the price of a basket of tomatoes rises from ₦5,000 to ₦6,000, the quantity supplied rises from 800 to 880 baskets. Find the PES and describe it. What PES would be expected immediately after harvest?

Solution: %ΔQs = 80/800 × 100 = 10%. %ΔP = 1,000/5,000 × 100 = 20%. PES = 10 ÷ 20 = 0.5, so supply is inelastic. Immediately after harvest, the stock is fixed, so PES is close to zero (perfectly inelastic in the market period).

Determinants of price elasticity of supply

1. **Time.** Supply is more elastic over longer periods because firms can build factories, plant trees or train workers. Economists distinguish the *market period* (supply fixed), the *short run* (some factors variable) and the *long run* (all factors variable).
2. **Availability of inputs.** If raw materials and labour are plentiful, output can be raised quickly.
3. **Spare capacity.** Factories with idle machines can increase output with little rise in cost.
4. **Ease of storing stock.** Durable products that can be stored (cement, canned goods) have more elastic supply than perishables (fresh fish).
5. **Nature of the product.** Manufactured goods can usually be expanded quickly; agricultural produce with a long growing period (cocoa, oil palm) cannot.
6. **Mobility of factors of production.** If resources can move easily from other industries, supply is elastic.
7. **Number of producers.** More firms means more potential response.

3.8 Elasticity and the incidence of taxation

When the government imposes an **indirect tax** such as value added tax or an excise duty, the legal burden falls on the seller, who adds the tax to the price. But who really bears the burden, the buyer or the seller? The division is called the **incidence of the tax**, and it depends on the elasticities of demand and supply. If demand is inelastic, consumers

cannot easily avoid the higher price, so most of the tax is passed on to them. If demand is elastic, consumers would cut purchases sharply, so sellers must absorb most of the tax. The same logic applies to supply: if supply is inelastic, producers cannot shift resources elsewhere and bear most of the burden.

WORKED EXAMPLE 3.7

A tax of ₦100 per unit is imposed on a product. The equilibrium price before the tax was ₦600; after the tax the market price paid by consumers is ₦680. Find the burden of the tax on consumers and producers.

Solution: Consumers pay ₦680 − ₦600 = ₦80 more per unit. The producer receives ₦680 − ₦100 = ₦580 per unit, which is ₦20 less than before. So consumers bear ₦80 of the ₦100 (80%) and producers bear ₦20 (20%). The fact that consumers bear more suggests that demand is relatively inelastic compared with supply.

3.9 Limitations and uses of elasticity

Elasticity values are estimates based on the past, and they change over time and between income groups. They assume ceteris paribus, which rarely holds in real life, since incomes, tastes and prices change together. Nevertheless, elasticity remains a powerful guide for firms setting prices, for governments choosing which goods to tax or subsidise, for forecasting the effects of exchange-rate changes on imports and exports

(the Marshall-Lerner condition says that a depreciation improves a country's trade balance only if the sum of the demand elasticities for exports and imports exceeds one), and for planning farm policy.

Exercise 3.1

1. Define price elasticity of demand and state the formula used to calculate it.
2. The price of a product rises from ₦200 to ₦250 and the quantity demanded falls from 400 to 340. Calculate PED and state whether demand is elastic or inelastic.
3. A 5% rise in the price of a good leads to a 15% fall in quantity demanded. Calculate PED and state the effect on total revenue of a price rise.
4. When price falls from ₦50 to ₦40, quantity demanded rises from 100 to 130. Calculate PED using the simple formula.
5. Income rises from ₦80,000 to ₦100,000 and demand for a good rises from 20 to 23 units. Calculate YED and classify the good.
6. A 10% rise in the price of tea causes a 12% rise in the demand for coffee. Calculate the cross elasticity and state the relationship.
7. The price of a good rises from ₦40 to ₦50 and the quantity supplied rises from 120 to 150. Calculate PES.
8. Explain why demand for salt is inelastic.
9. Why may a bumper harvest reduce farmers' incomes?

10. If PED = 0.4 and a firm raises its price by 10%, by how much does quantity demanded change and what happens to revenue?

Answers to Exercise 3.1

1. PED measures how responsive quantity demanded is to a price change: PED = %ΔQd ÷ %ΔP.
2. %ΔQ = −60/400 × 100 = −15%; %ΔP = 50/200 × 100 = 25%; PED = −15/25 = −0.6; inelastic.
3. PED = −15/5 = −3 (elastic). Total revenue falls when price rises.
4. %ΔQ = 30/100 × 100 = 30%; %ΔP = −10/50 × 100 = −20%; PED = −1.5 (absolute value 1.5).
5. %ΔY = 20,000/80,000 × 100 = 25%; %ΔQ = 3/20 × 100 = 15%; YED = 15/25 = 0.6; a necessity (normal, income-inelastic).
6. XED = 12/10 = +1.2; substitutes.
7. %ΔQ = 30/120 × 100 = 25%; %ΔP = 10/40 × 100 = 25%; PES = 1 (unitary).
8. It is a necessity, has no close substitutes and takes a tiny share of income.
9. Demand for food is inelastic, so a large rise in supply causes a big fall in price with only a small rise in quantity bought; revenue falls.
10. Quantity falls by 0.4 × 10 = 4%. Since demand is inelastic, revenue rises (price rises 10%, quantity falls 4%, so TR rises by about 5.6%: 1.10 × 0.96 = 1.056).

Exercise 3.2

1. A shop sells 500 pairs of shoes at ₦6,000 each. After a price cut to ₦5,000, it sells 700 pairs. (a) Calculate PED. (b) Compare total revenue before and after.
2. State the five categories of price elasticity of demand with the corresponding value ranges.
3. Use the midpoint formula to find PED if price rises from ₦90 to ₦110 and quantity falls from 220 to 180.
4. A good has YED = −0.5. What kind of good is it? Give a Nigerian example.
5. Explain why the demand for petrol is more elastic in the long run than in the short run.
6. A tax of ₦60 per unit raises the price paid by buyers from ₦300 to ₦345. What share of the tax is borne by consumers?
7. A supply curve is a straight line through the origin. What is PES at every point?
8. A firm's demand has PED = 2. It cuts price by 8%. Estimate the percentage change in quantity demanded and the direction of total revenue.
9. Why do governments tax cigarettes heavily?
10. The demand for printer ink cartridges falls by 6% when the price of printers rises by 15%. Calculate XED and interpret it.

Answers to Exercise 3.2

1. (a) %ΔQ = 200/500 × 100 = 40%; %ΔP = −1,000/6,000 × 100 = −16.67%; PED = −2.4 (absolute 2.4). (b) TR before = 6,000 × 500 = ₦3,000,000; after = 5,000 × 700 = ₦3,500,000; revenue rises by ₦500,000.
2. Perfectly inelastic (0); inelastic (0 to 1); unitary (1); elastic (above 1); perfectly elastic (infinity).
3. ΔQ = −40; average Q = 200; −40/200 = −0.2. ΔP = 20; average P = 100; 20/100 = 0.2. PED = −0.2/0.2 = −1, unitary.
4. An inferior good, such as very cheap second-hand clothing or low-grade rice.
5. Over time, drivers can switch to smaller cars, public transport or move nearer to work, so substitutes become available.
6. Buyers pay ₦45 more, so the share is 45/60 = 75%.
7. PES = 1 (unitary).
8. Quantity rises by 2 × 8 = 16%. Total revenue rises (1.16 × 0.92 = 1.0672, a rise of about 6.7%).
9. Demand is inelastic, so the tax raises large revenue without much fall in sales; it also discourages harmful consumption.
10. XED = −6/15 = −0.4; complements, with a weak relationship.

Chapter summary

- Elasticity measures responsiveness: it is the ratio of the percentage change in one variable to the percentage change in

another.

- PED = %ΔQd ÷ %ΔP. Demand is elastic if |PED| > 1, inelastic if below 1, and unitary at 1.
- Determinants of PED include substitutes, necessity, share of income, time, habit and width of definition.
- If demand is elastic, a price cut raises revenue; if inelastic, a price rise raises revenue.
- YED = %ΔQd ÷ %ΔIncome: positive for normal goods (above 1 for luxuries), negative for inferior goods.
- XED = %ΔQd of A ÷ %ΔP of B: positive for substitutes, negative for complements.
- PES = %ΔQs ÷ %ΔP; supply is more elastic in the long run, with spare capacity and storable goods.
- The burden of an indirect tax falls more on the side of the market that is less elastic.

Chapter 4: Theory of Production

Every loaf of bread in a Kaduna bakery, every bag of cement leaving Obajana, every barrel of crude oil pumped in the Niger Delta is the end product of a process in which resources are combined and transformed. **Production** is the creation of goods and services that satisfy human wants. It is not limited to making physical things: the work of a teacher, a bus driver and a bank clerk is production too, because it creates *utility*, the capacity to satisfy a want. Production is also not complete until the good reaches the person who will use it; a tomato in a Jos farm is of little use to a hungry resident of Lagos until it has been transported, stored and sold.

The theory of production asks practical questions. How should a firm combine labour, land and capital to produce as cheaply as possible? What happens to output when more workers are added to a fixed workshop? Why are some firms very large and others small? How are businesses organised? This chapter treats each of these questions and prepares the ground for the next chapter on costs and revenue.

4.1 Types and stages of production

Production is classified by what is done to resources. **Primary production** is the extraction of natural resources from land and water: farming, fishing, forestry, mining and oil drilling. **Secondary production** (manufacturing and construction) transforms raw materials

into finished or semi-finished goods: refining crude oil, milling flour, assembling vehicles, building houses. **Tertiary production** consists of services: trade, transport, banking, insurance, education, health, telecommunications and entertainment. Sometimes economists add a quaternary sector of knowledge-based services such as research and information technology.

Another distinction is between **direct production**, in which a person produces for his own consumption (a subsistence farmer growing cassava for his household), and **indirect production**, in which goods are made to be sold, so that producers buy what they need with the money earned. In modern economies indirect production dominates, and it requires specialisation and exchange.

Many products pass through successive stages. Cocoa beans are grown (primary), fermented and dried, then processed in a factory into cocoa powder and butter (secondary), packaged, transported to shops and sold (tertiary, called the *chain of production*). At each stage the product gains value; the extra value added at each stage is the difference between the value of output and the cost of inputs purchased from earlier stages.

4.2 Division of labour and specialisation

Division of labour is the breaking down of a production process into separate tasks, each performed by a different worker. **Specialisation** is the concentration of an individual, firm, region or country on the activity at which it is most efficient. A shirt factory in Aba may have one worker cutting cloth, another stitching sleeves, another attaching buttons and

another packing. Adam Smith's famous pin factory showed that ten workers dividing the tasks could produce thousands of times more pins than ten workers each making whole pins alone.

Advantages

- Workers become highly skilled at one task through repetition, so output rises.
- Time is saved because no one moves between tasks and tools.
- Machines can be introduced to do simple repetitive tasks, raising productivity.
- Workers can be matched to jobs according to their talents.
- Mass production lowers the cost per unit and so lowers prices for consumers.
- Training time is short, because workers learn a narrow task quickly.

Disadvantages

- Repetitive work is monotonous and may cause boredom and low morale.
- Workers lose all-round skills; if their task disappears, they find it hard to move to another job (occupational immobility).
- Interdependence is a danger: a strike or breakdown at one stage halts the whole line.
- Standardised products offer less variety and little pride of craftsmanship.

- Over-specialisation by regions or countries makes them vulnerable to price falls in one product. Nigeria's heavy dependence on crude oil is an example.

Key point. The extent of division of labour is limited by the size of the market. A small village market cannot support a shop that sells only one type of watch, but a large city can. This is why division of labour is more advanced in Lagos than in a small rural town.

4.3 Production in the short run and the long run

Economists define time by flexibility rather than by the calendar. In the **short run** at least one factor of production, typically capital or land, is fixed, so that output can be changed only by varying the other factors, such as labour or raw materials. In the **long run** all factors are variable, so a firm can build a new factory, buy new machines and change its scale. Costs and production decisions differ greatly between the two periods.

4.4 Total, average and marginal product

Consider a small bakery with one oven (fixed factor) and a variable number of workers. Three measures describe how output responds as workers are added:

- **Total product (TP)**: the total output produced by all workers.
- **Average product (AP)**: output per worker, AP = TP ÷ number of workers.
- **Marginal product (MP)**: the addition to total product from employing one more worker, MP = ΔTP ÷ ΔL (change in total product divided by change in labour).

The table shows a bakery's output of loaves per day.

Workers (L)	Total product (TP)	Average product (AP)	Marginal product (MP)
0	0	–	–
1	10	10	10
2	24	12	14
3	39	13	15
4	52	13	13
5	60	12	8
6	64	10.67	4
7	63	9	−1

Observe the pattern. At first, MP rises as workers share tasks and use the oven more fully. After the third worker, MP begins to fall: the fourth adds 13, the fifth 8, the sixth only 4, and the seventh actually reduces output because the kitchen is overcrowded. TP rises at first at an increasing rate, then at a decreasing rate, reaches its maximum when MP is zero, and falls when MP is negative.

Key point. The relationship between AP and MP: when MP is above AP, AP is rising; when MP is below AP, AP is falling; MP cuts AP at the maximum of AP. In the table, MP (15) exceeds AP (12) when the third worker joins, so AP rises to 13; the fourth worker has MP = 13 = AP, so AP is at its maximum; afterwards MP is below AP and AP falls.

WORKED EXAMPLE 4.1

A farm in Benue uses a fixed plot of land. The total output of maize (in bags) for 1 to 5 workers is 8, 20, 36, 44 and 48. Calculate the marginal and average products and identify where diminishing returns begin.

Solution: MP: 1st worker 8; 2nd 20 − 8 = 12; 3rd 36 − 20 = 16; 4th 44 − 36 = 8; 5th 48 − 44 = 4. AP: 8/1 = 8; 20/2 = 10; 36/3 = 12; 44/4 = 11; 48/5 = 9.6. MP rises until the third worker (16) and falls from the fourth worker (8). Diminishing returns therefore begin with the fourth worker. AP peaks at 12 with three workers, and MP (8) is below AP (11) at the fourth worker, so AP falls.

4.5 The law of diminishing returns

The **law of diminishing returns** (also called the law of variable proportions) states that, in the short run, as additional units of a variable factor are combined with a fixed factor, the marginal product of the

variable factor will eventually decline. Notice the key words: *eventually*, *short run* and *fixed factor*. The law does not say that output falls; it says the *extra* output from each new unit of the variable factor falls after a point. It also assumes the quality of the variable factor and the state of technology are unchanged.

The law explains why a fixed amount of land cannot feed unlimited numbers of farmers. If ten farmers work an acre, adding an eleventh does not add as much as the tenth did. The law is also the main reason why short-run marginal cost eventually rises, as we shall see in Chapter 5. The three **stages of production** are:

- **Stage I (increasing returns)**: from the origin to the point where AP is highest; MP rises and then starts to fall but remains above AP. A rational firm will not stop here, since each extra worker raises the average output.
- **Stage II (diminishing returns)**: from maximum AP to the point where MP is zero (TP maximum). The rational firm operates in this stage.
- **Stage III (negative returns)**: MP is negative and TP is falling. No rational firm would operate here, since it could produce more with fewer workers.

4.6 Returns to scale

The long run allows all factors to change together. **Returns to scale** describe how output responds when all inputs are increased in the same proportion. If doubling all inputs more than doubles output, there are **increasing returns to scale**. If output exactly doubles, **constant returns**. If output less than doubles, **decreasing returns**. Do not confuse this long-run concept with diminishing returns, a short-run idea.

WORKED EXAMPLE 4.2

A factory uses 10 workers and 5 machines to make 400 units. When it doubles to 20 workers and 10 machines, output becomes 900 units. When it doubles again to 40 workers and 20 machines, output is 1,800 units. Identify the returns to scale in each case.

Solution: First doubling: inputs ×2 and output rises from 400 to 900, which is ×2.25, more than double, so increasing returns to scale. Second doubling: inputs ×2 and output goes from 900 to 1,800, which is exactly ×2, so constant returns to scale.

4.7 Economies and diseconomies of scale

As a firm grows larger in the long run, its average cost of production per unit may fall; these cost advantages are **economies of scale**. They are of two main types.

Internal economies of scale

These arise from the growth of the firm itself.

- **Technical economies.** Large firms can use big, specialised machinery and mass production. A big bakery can use an automated oven unaffordable for a small shop. Increased dimensions also help: doubling the size of a tanker more than doubles its capacity.
- **Managerial economies.** Large firms can employ specialist managers for finance, marketing and production, which improves efficiency.
- **Financial economies.** Big firms borrow more cheaply, because banks regard them as safe, and can raise capital by selling shares.
- **Marketing economies.** Buying raw materials in bulk attracts discounts; advertising costs are spread over a bigger output; distribution is cheaper per unit.
- **Risk-bearing economies.** Diversification into several products or markets reduces the danger that a single failure will ruin the firm.
- **Welfare economies.** Large firms provide canteens, clinics and housing, which raise worker morale and productivity.

External economies of scale

These come from the growth of the whole industry, not the single firm, and are available to all firms in a locality. They include a skilled labour pool, specialised suppliers, shared transport and communication networks, and training colleges. Examples are the cluster of electronics traders at Ikeja Computer Village and the textile industry formerly

concentrated in Kano. The concentration of an industry in a particular place is called **localisation of industry**. Its disadvantages include congestion, traffic, higher land costs, and vulnerability of the region if the industry declines.

Diseconomies of scale

Beyond some size, average cost per unit may start to rise. Internal causes include management problems (communication delays, bureaucracy, loss of control), poor worker motivation in a huge organisation, and conflict between departments. External diseconomies include traffic congestion, pollution, higher wages and rents caused by too many firms in one area. The size of output at which average cost is lowest is called the **optimum size** or minimum efficient scale of the firm.

4.8 Factors determining the size of a firm

The size of a firm, measured by employment, output or capital, depends on the extent of the market, the nature of the product, the availability of capital, the quality of management, government policy and the type of ownership. Where a product is standardised and demand is large (cement, beer), large firms predominate. Where the service is personal and local (hairdressing, tailoring, small food stalls), small firms survive, because customers want individual attention and the market is limited. Firms also grow by **merger** (two firms agree to combine) and **take-over** (one firm buys another). A **horizontal** merger joins firms at the same

stage of production (two banks); a **vertical** merger joins firms at different stages (a flour miller and a bakery); a **conglomerate** merger joins firms in unrelated industries.

Small firms survive in the economy because of limited markets, personal service, low capital needs, the ability to meet specialised demand, flexible decision-making and the government's support for small and medium enterprises. In Nigeria, small and medium enterprises employ a large part of the workforce.

4.9 Productivity

Productivity is output per unit of input. Labour productivity is output per worker (or per hour); it is a key determinant of living standards. Productivity rises with education and training, better technology, more capital per worker, good health, sound management, and an efficient infrastructure. Low productivity in Nigeria is often traced to power shortages, poor roads and weak skills.

WORKED EXAMPLE 4.3

A garment factory employs 40 workers and produces 1,200 shirts per week. After training and the purchase of new sewing machines, it employs 45 workers and produces 1,980 shirts. Find the percentage change in labour productivity.

Solution: Initial productivity = 1,200/40 = 30 shirts per worker. New productivity = 1,980/45 = 44 shirts per worker. Increase = 44 − 30 = 14; percentage change = 14/30 × 100 = 46.67%.

4.10 Forms of business organisation

Firms are owned and controlled in several ways, each with its own advantages and drawbacks.

Sole proprietorship

One person owns and usually manages the business, as with a roadside provision store or a tailoring shop. Advantages: easy to start, full control, all profit retained, quick decisions, simple secrecy. Disadvantages: limited capital, **unlimited liability** (the owner's personal property can be seized to pay debts), limited skills, and the business ends with the owner's death or retirement.

Partnership

Two to twenty persons (the traditional legal ceiling for an ordinary business partnership, with exceptions for some professional firms) own the business together under a deed or agreement. Advantages: more capital and skills, shared risk. Disadvantages: unlimited liability for general partners, disagreements, each partner's acts bind the others, and the firm may dissolve on a partner's death or withdrawal. A *limited partner* has liability limited to his contribution but may not take part in management.

Limited liability companies

A company is a legal person, separate from its owners (shareholders), registered with the Corporate Affairs Commission. Shareholders' liability is limited to the value of their shares. A **private limited company** (Ltd) restricts the transfer of shares and may not invite the public to buy them. A **public limited company** (Plc) may sell shares to the public, usually through the stock exchange, and so can raise very large capital. Advantages: limited liability, perpetual succession, large capital, specialised management. Disadvantages: complex formation and legal rules, separation of ownership from control, publication of accounts, and taxation of profits. Shareholders receive **dividends**, the portion of profit distributed to them.

Cooperative societies

These are voluntary associations of people with a common need who pool resources for mutual benefit, such as farmers' marketing cooperatives, thrift and credit societies and consumer cooperatives. Principles include voluntary membership, democratic control (one member, one vote) and the sharing of surplus in proportion to patronage. Their weaknesses are limited capital and management skill, and sometimes mismanagement and dishonesty.

Public enterprises

Public corporations are owned by the government, set up by an act or decree to supply essential services (the Nigerian Ports Authority, Nigerian Railway Corporation). The reasons for establishing them include providing essential services, controlling natural monopolies, safeguarding national security and encouraging balanced regional development. Their problems include inefficiency, political interference, corruption, losses requiring government subsidy, and lack of competition. Governments have turned to **privatisation** (selling state enterprises to private investors) and **commercialisation** (requiring them to run as profit-making businesses while remaining in government hands) to correct these. The opposite, taking private firms into state ownership, is **nationalisation**. In the Nigerian context, **indigenisation** means the policy, begun in the 1970s, of requiring that certain foreign-owned businesses be transferred wholly or partly to Nigerian ownership.

WORKED EXAMPLE 4.4

Mr. Adeyemi and Ms. Bello start a partnership to run a bakery in Ibadan. Adeyemi contributes ₦3,000,000 and Bello ₦2,000,000. They agree to share profits in proportion to capital. The profit for the year is ₦1,500,000. If the business later owes ₦9,000,000 and has assets of only ₦4,000,000, explain what happens under unlimited liability.

Solution: Total capital = ₦5,000,000. Adeyemi's share = 3/5 × 1,500,000 = ₦900,000; Bello's share = 2/5 × 1,500,000 = ₦600,000. The shortfall in the later case is ₦9,000,000 − ₦4,000,000 = ₦5,000,000. Because liability is unlimited, creditors can claim the partners' personal property to settle the ₦5,000,000, and each partner can be held responsible for the whole debt. If the business were a limited company, the shareholders' loss would be limited to the money they put into the shares.

Exercise 4.1

1. Distinguish between primary, secondary and tertiary production with one Nigerian example each.
2. State three advantages and three disadvantages of division of labour.
3. Define the law of diminishing returns and state its assumptions.

4. Complete the table for a farm: workers 1, 2, 3, 4, 5 with total product 6, 18, 33, 44, 50. Find MP and AP for each.
5. In Question 4, at which worker do diminishing returns begin? Between which workers does AP reach a maximum?
6. A firm doubles all inputs and output rises by 60%. What type of returns to scale is this?
7. Explain two internal economies of scale.
8. Distinguish between internal and external economies of scale.
9. Output per worker in a firm rises from 25 units to 30 units. Calculate the percentage increase in productivity.
10. State two advantages and two disadvantages of a sole proprietorship.

Answers to Exercise 4.1

1. Primary: extraction (cocoa farming in Ondo, crude oil in Bayelsa); secondary: manufacturing (cement at Obajana); tertiary: services (banking, transport).
2. Advantages: skill, time saving, mass production, lower unit cost. Disadvantages: monotony, immobility, interdependence, vulnerability of specialised regions.
3. As more units of a variable factor are added to a fixed factor in the short run, MP of the variable factor eventually falls. Assumes at least one fixed factor, constant technology and equal quality of the variable factor.

4. MP: 6, 12, 15, 11, 6. AP: 6, 9, 11, 11, 10. (TP: 18 − 6 = 12; 33 − 18 = 15; 44 − 33 = 11; 50 − 44 = 6. AP: 18/2 = 9; 33/3 = 11; 44/4 = 11; 50/5 = 10.)
5. MP is highest at the third worker (15), so diminishing returns begin with the fourth worker. AP is the same (11) at 3 and 4 workers; its maximum is reached at the fourth worker, where MP = AP = 11.
6. Output rose by 60% while inputs rose by 100%, so decreasing returns to scale.
7. Technical (large machines, mass production) and managerial (specialist managers); also financial, marketing, risk-bearing.
8. Internal economies arise from the growth of the firm itself; external economies arise from the growth of the industry and are shared by all firms.
9. (30 − 25)/25 × 100 = 20%.
10. Advantages: easy to form; owner keeps all profit. Disadvantages: unlimited liability; limited capital.

Exercise 4.2

1. The total product of a firm with 1 to 6 workers is 5, 14, 27, 36, 40, 40. Find the marginal product of each worker and state the worker beyond which total product stops rising.
2. Explain why a rational firm will not operate in Stage III of production.
3. Output is 500 units with 20 workers. When labour rises to 25, output rises to 575. Calculate the MP of the extra workers and

the AP before and after.

4. Explain why the existence of external economies of scale leads firms in the same industry to cluster in one area.
5. State four problems of public enterprises in Nigeria.
6. Differentiate between a private limited company and a public limited company.
7. A company has a profit of ₦8,000,000 and 4,000,000 shares. If it distributes ₦5,000,000 as dividends, find the dividend per share.
8. Differentiate between privatisation and commercialisation.
9. Why does the size of the market limit the division of labour?
10. Three partners contribute ₦4m, ₦3m and ₦1m and share profit in proportion to capital. Find each partner's share of a profit of ₦960,000.

Answers to Exercise 4.2

1. MP: 5, 9, 13, 9, 4, 0. TP stops rising after the fifth worker; the sixth adds nothing.
2. In Stage III MP is negative; total output could be increased by employing fewer workers, saving wages as well.
3. MP = 75/5 = 15 per worker. AP before = 500/20 = 25; AP after = 575/25 = 23.
4. Firms benefit from skilled labour, suppliers and infrastructure that exist because other firms are there, reducing costs.

5. Inefficiency, political interference, corruption, heavy losses/subsidies, lack of competition, overstaffing.
6. A private company restricts share transfer and cannot sell shares to the public; a public company may offer shares to the public and be listed on the stock exchange.
7. 5,000,000/4,000,000 = ₦1.25 per share.
8. Privatisation transfers ownership of state enterprises to private owners; commercialisation keeps government ownership but requires profit-oriented management.
9. Specialisation produces large output, which needs many buyers; if the market is small, the extra output cannot be sold.
10. Total = ₦8m. ₦4m: 4/8 × 960,000 = ₦480,000; ₦3m: 3/8 × 960,000 = ₦360,000; ₦1m: 1/8 × 960,000 = ₦120,000.

Chapter summary

- Production creates utility. It is classified as primary, secondary and tertiary, and takes place through successive stages in which value is added.
- Division of labour raises output by skill and time saving, but causes monotony, immobility and interdependence; it is limited by the size of the market.
- In the short run at least one factor is fixed. MP = ΔTP/ΔL, AP = TP/L; MP cuts AP at its maximum.
- The law of diminishing returns says that the marginal product of a variable factor eventually falls when it is added to a fixed

factor. Firms operate in Stage II.

- Returns to scale is a long-run concept: increasing, constant and decreasing returns depend on how output compares with the proportional rise in all inputs.
- Economies of scale (technical, managerial, financial, marketing, risk-bearing) lower average cost; external economies come from industry growth. Diseconomies raise average cost.
- Productivity is output per unit of input and depends on skills, technology and infrastructure.
- Forms of business include sole proprietorship, partnership, companies, cooperatives and public enterprises, differing in liability, capital and control.

Chapter 5: Costs, Revenue and Profit

A woman in Aba who sews school uniforms must decide how many to make each week. She knows what she charges, and she knows that each extra uniform needs cloth, thread and some of her time. But she also pays rent on her shop whether she sews one uniform or a hundred, and she must pay her electricity bill, her apprentice's wages and the instalments on her machine. To know whether her business is succeeding, and how much she should produce, she needs a clear picture of her **costs** and her **revenue**. The difference between them is her **profit**.

The theory of costs and revenue is the bridge between production (Chapter 4) and the study of market structures (Chapter 6). It explains why a firm's supply curve slopes upward, why some firms shut down and others expand, and why the output that maximises profit is the one at which marginal cost equals marginal revenue. In this chapter we define each cost and revenue concept, show how to calculate it from a table, explain the shapes of the cost curves, and apply the profit-maximising rule.

5.1 The meaning of cost

In economics, the **cost of production** is the payment needed to obtain all the factors of production used to produce a good. Economists treat cost more broadly than accountants do. **Explicit costs** are payments

actually made to outsiders: wages, rent, purchased materials, electricity. **Implicit costs** are the opportunity costs of resources the owner already owns and supplies himself: the salary he could have earned elsewhere, the rent he could have charged for his own building, the interest on his own capital. **Accounting profit** subtracts only explicit costs from revenue; **economic profit** subtracts both explicit and implicit costs.

Normal profit is the minimum reward an entrepreneur must earn to remain in his present business. It is treated as an implicit cost, so when a firm earns just normal profit, its economic profit is zero. **Supernormal profit** (abnormal or excess profit) is any profit above normal profit. A firm earning less than normal profit is making a **subnormal profit** (economic loss) and has an incentive to leave the industry in the long run.

5.2 Short-run costs

In the short run, at least one factor is fixed, so costs fall into two groups.

- **Fixed costs (FC)** do not vary with output in the short run. They must be paid even if output is zero. Examples: rent, insurance, interest on loans, salaries of permanent managers, depreciation of buildings.
- **Variable costs (VC)** change with output. Examples: raw materials, fuel, wages of casual workers, packaging.

From these we define:

- **Total cost (TC)** = Total fixed cost (TFC) + Total variable cost (TVC).
- **Average fixed cost (AFC)** = TFC ÷ Q. It falls continuously as output rises, since the same fixed sum is spread over more units. This is called "spreading the overhead."
- **Average variable cost (AVC)** = TVC ÷ Q.
- **Average (total) cost (ATC or AC)** = TC ÷ Q = AFC + AVC.
- **Marginal cost (MC)** = the addition to total cost from producing one more unit = ΔTC ÷ ΔQ. Because fixed cost does not change, MC is also ΔTVC ÷ ΔQ.

Consider a small block-making firm with fixed costs of ₦60 (in thousands of naira) per week. The table shows its costs.

Output Q	TFC	TVC	TC	AFC	AVC	ATC	MC
0	60	0	60	–	–	–	–
1	60	40	100	60	40	100	40
2	60	70	130	30	35	65	30
3	60	90	150	20	30	50	20
4	60	120	180	15	30	45	30
5	60	170	230	12	34	46	50
6	60	240	300	10	40	50	70

WORKED EXAMPLE 5.1

Verify the row for Q = 5 in the table, and find the marginal cost of increasing output from 4 to 5 units.

Solution: TFC = 60 and TVC = 170, so TC = 60 + 170 = 230. AFC = 60/5 = 12; AVC = 170/5 = 34; ATC = 230/5 = 46 (also 12 + 34 = 46). MC = TC(5) − TC(4) = 230 − 180 = 50.

Shapes of the cost curves

AFC slopes down continuously. AVC, ATC and MC are **U-shaped**. The reason lies in the law of diminishing returns. At low output, as extra workers are added, their marginal product rises, so extra output costs less and less; MC falls. When diminishing returns set in, each extra unit of output requires more and more extra input, so MC rises. AVC falls while MC is below it and rises when MC is above it. ATC does the same, with the added effect of falling AFC, so ATC reaches its minimum at a larger output than AVC.

Key point. The marginal cost curve cuts both the AVC and the ATC curves at their lowest points. When MC is below the average, the average falls; when MC is above the average, the average rises. In the table, ATC is lowest (45) at Q = 4, where MC (30) has not yet risen above it; at Q = 5, MC (50) is greater than ATC, so ATC rises to 46. The true crossing lies between Q = 4 and Q = 5.

5.3 Long-run costs

In the long run all costs are variable. The **long-run average cost (LRAC)** curve shows the lowest cost per unit of producing each level of output when the firm can choose any scale of plant. It is derived as an "envelope" of many short-run average cost curves, each for a different plant size, and it is U-shaped (or L-shaped) because of economies and diseconomies of scale. The falling section reflects economies of scale, the lowest point the optimum scale, and the rising section diseconomies of scale.

5.4 The concept of revenue

Revenue is the money a firm receives from selling its output.

- **Total revenue (TR)** = price × quantity sold = P × Q.
- **Average revenue (AR)** = TR ÷ Q = price. The AR curve is the demand curve facing the firm.
- **Marginal revenue (MR)** = the addition to total revenue from selling one more unit = ΔTR ÷ ΔQ.

The shape of the revenue curves depends on the market structure. In **perfect competition** the firm is a price taker: it can sell any quantity at the market price. Every extra unit adds exactly the price to revenue, so AR = MR = price, a horizontal line, and TR rises as a straight line from the origin. In **imperfect competition** (monopoly, oligopoly, monopolistic competition) the firm faces a downward-sloping demand

curve: to sell more it must lower the price, and in most cases it lowers the price on all units, so MR is below AR, and the MR curve lies below the AR curve. For a straight-line demand curve, the MR curve has the same intercept on the price axis but twice the slope, so it cuts the quantity axis at half the distance of the demand curve. TR rises while MR is positive, is at a maximum when MR = 0 (demand has unitary elasticity at that point), and falls when MR is negative.

WORKED EXAMPLE 5.2

A monopolist faces the demand schedule: price ₦90, ₦80, ₦70, ₦60, ₦50, ₦40 for quantities 1, 2, 3, 4, 5, 6. Calculate TR, AR and MR, and find the output at which TR is maximum.

Solution: TR = P × Q: 90, 160, 210, 240, 250, 240. AR = price: 90, 80, 70, 60, 50, 40. MR = change in TR: 90, 70, 50, 30, 10, −10. TR is maximum (250) at Q = 5, where MR (10) is just above zero; at Q = 6 MR is negative, so TR falls to 240.

5.5 Profit and the profit-maximising rule

Profit = total revenue − total cost. A firm that aims to maximise profit should keep expanding output as long as the extra unit adds more to revenue than to cost, that is, as long as MR > MC. It should stop when producing another unit would add more to cost than revenue (MR < MC). So profit is maximised where:

MC = MR, with the MC curve cutting the MR curve from below (that is, MC rising).

This rule applies to every market structure. In perfect competition MR = P, so the rule becomes MC = P.

WORKED EXAMPLE 5.3

The block-making firm of the table in Section 5.2 is a perfect competitor and can sell at ₦60 per unit. Find the profit-maximising output and the profit.

Solution: MR = P = 60. MC values for units 1 to 6 are 40, 30, 20, 30, 50, 70. MC is below MR up to the fifth unit (50 < 60) but exceeds it at the sixth (70 > 60). Output = 5. TR = 60 × 5 = 300; TC = 230; profit = 300 − 230 = ₦70 thousand. Check: at Q = 4, profit = 240 − 180 = 60; at Q = 6, 360 − 300 = 60. Both are below 70, so Q = 5 is best.

WORKED EXAMPLE 5.4

The same cost schedule applies, but the firm is now a monopolist facing the demand schedule of Worked example 5.2 (MR = 90, 70, 50, 30, 10, −10). Find the profit-maximising output, the price and the profit.

Solution: Compare MR and MC: Q = 1: 90 > 40; Q = 2: 70 > 30; Q = 3: 50 > 20; Q = 4: 30 = 30; Q = 5: 10 < 50. Output is 4 (where MR = MC). The price from the demand schedule is ₦60. TR = 60 × 4 = 240; TC = 180; profit = ₦60 thousand. The firm does not produce the fifth unit because it would add 10 to revenue but 50 to cost.

5.6 Break-even, shutdown and long-run exit

A firm is at its **break-even point** when total revenue equals total cost, so that profit is zero in the accounting sense (and the owner earns only normal profit in the economic sense when implicit costs are included). For a perfectly competitive firm this occurs where the price equals the minimum of ATC. In the table the minimum ATC is about ₦45 at Q = 4, so a market price of ₦45 gives TR = 180 = TC and the firm breaks even.

Consider what happens when price falls below ATC. The firm makes a loss, but should it close at once? In the **short run**, the answer depends on AVC. If price exceeds AVC, the firm covers all its variable costs and part of its fixed costs; closing down would lose the whole fixed cost, so it is better to carry on producing. If price falls below AVC, each unit

sold loses money even before fixed costs, so the firm should shut down immediately and lose only its fixed cost. The lowest point of AVC is therefore the **shutdown point**, and the part of the MC curve above AVC is the firm's short-run supply curve in perfect competition. In the **long run**, a firm that cannot cover ATC will leave the industry, since fixed costs can be avoided by selling the equipment.

Exam tip. Summary of price and cost comparisons for a perfectly competitive firm: P > ATC, supernormal profit; P = ATC, normal profit (break-even); AVC < P < ATC, loss but continue in the short run; P < AVC, shut down.

WORKED EXAMPLE 5.5

A firm has total fixed cost of ₦20,000 per month. At an output of 500 units, total variable cost is ₦35,000. The market price is ₦65. Should the firm continue in the short run?

Solution: AVC = 35,000/500 = ₦70; ATC = (20,000 + 35,000)/500 = 55,000/500 = ₦110. Price (65) is below AVC (70), so each unit sold loses ₦5 before fixed costs. Total revenue = 65 × 500 = ₦32,500, which is less than TVC of ₦35,000. The firm should shut down: by closing, it loses only ₦20,000 of fixed costs, whereas producing loses 55,000 − 32,500 = ₦22,500.

5.7 Calculating costs from equations

Sometimes the total cost is given by a formula. Marginal cost then follows by applying the definition to successive units. For example, if $TC = 100 + 10Q + Q^2$, then $TFC = 100$, $TVC = 10Q + Q^2$, $AFC = 100/Q$, $AVC = 10 + Q$, $ATC = 100/Q + 10 + Q$. For output of 8, $TC = 100 + 80 + 64 = 244$, and $ATC = 244/8 = 30.5$. The marginal cost of the 8th unit is $TC(8) - TC(7) = 244 - (100 + 70 + 49) = 244 - 219 = 25$. (For the continuous version, $MC = 10 + 2Q$, which gives 26 at $Q = 8$; examination questions that give tables use the difference method.)

WORKED EXAMPLE 5.6

A firm's total cost is $TC = 50 + 6Q + Q^2$ (in thousands of naira). Find the average cost and the marginal cost of the 5th unit.

Solution: $TC(5) = 50 + 30 + 25 = 105$. $ATC = 105/5 = 21$. $TC(4) = 50 + 24 + 16 = 90$. MC of the 5th unit $= 105 - 90 = 15$. The average cost is ₦21 thousand and the marginal cost ₦15 thousand. Since MC is below ATC, average cost is still falling.

5.8 Why average cost matters to price and entry

When firms in an industry are making supernormal profit, new firms are attracted; supply increases, price falls, and profits shrink. When firms make losses, firms leave, supply decreases, price rises. In the long run in

perfect competition, entry and exit force price toward the minimum of long-run average cost, so firms earn only normal profit. In markets with barriers to entry such as patents or high capital requirements, supernormal profits may persist. The structure of costs therefore helps determine how many firms survive and how efficiently they operate.

Exercise 5.1

1. Distinguish between fixed and variable costs, giving two examples of each.
2. Distinguish between explicit and implicit costs, and between accounting profit and economic profit.
3. A firm has TFC = ₦12,000. At 400 units, TVC = ₦28,000. Calculate TC, AFC, AVC and ATC.
4. Total cost is ₦500 at 10 units and ₦540 at 11 units. Find the marginal cost of the 11th unit.
5. Complete: Q = 1, 2, 3, 4 with TC = 55, 70, 80, 100 and TFC = 40. Find MC and AVC at each level.
6. Define marginal revenue. Why does MR equal price in perfect competition?
7. A firm sells 20 units at ₦50 each and then cuts the price to ₦45 to sell 24 units. Calculate TR in each case and the marginal revenue per extra unit.
8. State the profit-maximising rule and explain it.
9. Why does a firm continue in production in the short run if price is above AVC but below ATC?

10. Why does AFC continuously fall as output rises?

Answers to Exercise 5.1

1. Fixed costs do not change with output (rent, insurance); variable costs do (raw materials, casual wages).
2. Explicit costs are actual payments to outsiders; implicit costs are opportunity costs of owner-supplied resources. Accounting profit = revenue − explicit costs; economic profit = revenue − explicit and implicit costs.
3. TC = 12,000 + 28,000 = ₦40,000; AFC = 12,000/400 = ₦30; AVC = 28,000/400 = ₦70; ATC = 40,000/400 = ₦100.
4. MC = 540 − 500 = ₦40.
5. MC: Q1: 55 − 40 = 15 (TC at Q0 = TFC = 40); Q2: 70 − 55 = 15; Q3: 80 − 70 = 10; Q4: 100 − 80 = 20. TVC = TC − 40 = 15, 30, 40, 60; AVC = 15, 15, 13.33, 15.
6. MR is the addition to TR from selling one more unit. In perfect competition each extra unit sells at the same market price, so it adds the price to revenue.
7. TR(20) = ₦1,000; TR(24) = 45 × 24 = ₦1,080. Extra revenue = ₦80 for 4 extra units, so MR = ₦20 per unit on average.
8. Produce where MC = MR (MC rising): up to that point, each extra unit adds more to revenue than to cost.
9. It covers all variable costs and part of fixed costs; shutting down would lose all fixed costs.

10. The fixed sum is spread over more units.

Exercise 5.2

1. A perfectly competitive firm faces a price of ₦40. Its MC for units 1 to 6 is 15, 20, 28, 36, 45, 58. What output maximises profit?
2. If the TFC of the firm in Question 1 is ₦30 and TVC for 4 units is ₦99, calculate its profit.
3. TC = 200 + 4Q + Q^2. Find TC and ATC at Q = 10, and the cost of the 10th unit as TC(10) − TC(9).
4. A monopolist's demand: Q = 1 → P = 100; Q = 2 → 90; Q = 3 → 80; Q = 4 → 70. Calculate TR and MR.
5. If the monopolist in Question 4 has MC = 55 constant, what output maximises profit?
6. Price is ₦25, AVC ₦30 and ATC ₦45. Should the competitive firm continue in the short run? Why?
7. Explain why MC cuts ATC at its minimum.
8. State two reasons why new firms enter an industry that is earning supernormal profit.
9. Output is 300 units; TR = ₦18,000; TC = ₦15,300. Calculate price, profit and profit per unit.

Answers to Exercise 5.2

1. MC < 40 for units 1 to 4 (36 < 40) but unit 5 has MC 45 > 40. Output = 4.
2. TR = 40 × 4 = 160; TC = 30 + 99 = 129; profit = ₦31.
3. TC(10) = 200 + 40 + 100 = 340; ATC = 34. TC(9) = 200 + 36 + 81 = 317; MC = 340 − 317 = 23.
4. TR = 100, 180, 240, 280. MR = 100, 80, 60, 40.
5. MR exceeds 55 for the first three units (100, 80, 60) and falls to 40 for the fourth. Output = 3; price = ₦80.
6. No. Price (25) is below AVC (30): it does not even cover variable costs, so it should shut down.
7. When MC is below ATC, producing another unit pulls the average down; when above, it pulls it up. So ATC stops falling and begins rising exactly where MC equals it.
8. The chance of earning above normal profit; low barriers to entry (and resource mobility).
9. Price = 18,000/300 = ₦60; profit = 18,000 − 15,300 = ₦2,700; per unit = 2,700/300 = ₦9.

Chapter summary

- Costs include explicit and implicit items. Normal profit is part of economic cost; supernormal profit exceeds it.
- TC = TFC + TVC; AFC = TFC/Q; AVC = TVC/Q; ATC = AFC + AVC; MC = ΔTC/ΔQ.

- AVC, ATC and MC are U-shaped because of diminishing returns, and MC cuts AVC and ATC at their minimum points.
- TR = P × Q; AR = price; MR = ΔTR/ΔQ. In perfect competition AR = MR = P; under imperfect competition MR lies below AR.
- Profit is maximised where MC = MR with MC rising.
- A firm breaks even where TR = TC (P = ATC). It shuts down in the short run if P < AVC, and leaves the industry in the long run if P < ATC.
- Long-run average cost is U-shaped because of economies and diseconomies of scale.

Chapter 6: Market Structures

Walk through any Nigerian city and you will meet markets of very different kinds. At Mile 12 in Lagos or Bodija in Ibadan, hundreds of traders sell tomatoes, pepper and onions that look almost identical, and no single trader can set the price. Pay your electricity bill and you deal with a distribution company that is the only supplier in your area. Buy a mobile data plan and you choose among only a handful of large networks that watch one another's prices closely. Choose a hairdresser or a tailor and you find many sellers, each offering a slightly different style, reputation and location. These are not accidents. Each market has a different **structure**, and the structure determines how firms set prices, how much they produce, how much profit they earn, and how well consumers are served.

In Chapter 5 we learnt that every firm maximises profit where marginal cost equals marginal revenue. In this chapter we ask what the revenue curves look like in each type of market and what follows. We study perfect competition, monopoly, monopolistic competition and oligopoly, compare their results, and consider the ways in which governments regulate firms with market power.

6.1 What is a market structure?

A **market** is any arrangement that brings buyers and sellers of a good or service into contact, whether in a physical place or over the telephone or internet. A **market structure** is the set of features of a market that determines how its firms behave. The main features are:

- the **number and size** of the firms, and of the buyers;
- whether the products are **homogeneous** (identical) or **differentiated**;
- the **freedom of entry and exit**, that is, the barriers that prevent new firms from joining;
- the quality of **information** that buyers and sellers possess; and
- the degree of **control over price** that an individual firm has.

Economists place markets on a spectrum. At one end is **perfect competition**, where firms have no control over price at all. At the other end is **pure monopoly**, where a single firm controls the market. Between them lie **monopolistic competition** and **oligopoly**. Everything between perfect competition and pure monopoly is called **imperfect competition**.

Feature	Perfect competition	Monopolistic competition	Oligopoly	Monopoly
Number of sellers	Very many	Many	Few	One
Product	Homogeneous	Differentiated	Homogeneous or differentiated	Unique, no close substitute
Entry and exit	Free	Fairly free	Restricted	Blocked
Control over price	None (price taker)	Slight	Considerable but interdependent	Considerable (price maker)
Demand curve faced by the firm	Horizontal	Downward-sloping, fairly elastic	Kinked or uncertain	Downward-sloping, market demand
Long-run profit	Normal	Normal	Often supernormal	Often supernormal

6.2 Perfect competition

Characteristics

Perfect competition is an ideal market, rarely met in full, but it is the standard against which other markets are judged. It requires the following conditions.

1. **A very large number of buyers and sellers**, each so small relative to the market that no single one can affect the price by changing

output.

2. **Homogeneous products.** The goods of all sellers are identical, so buyers have no reason to prefer one seller to another. Brands and advertising play no part.
3. **Freedom of entry and exit.** New firms can enter and unprofitable firms can leave without legal or financial barriers.
4. **Perfect knowledge.** Buyers and sellers know all prices, and all producers know the best available techniques.
5. **Perfect mobility of factors of production** between uses and places.
6. **No government intervention** and no transport cost differences that separate the market.

Markets that come close include the markets for staple farm produce such as maize, yam, garri and tomatoes, and for foreign exchange and shares on an organised exchange. Even there, real life departs from the ideal: some traders form associations, perishable goods vary in quality, and information is not complete.

The price taker

Because each firm is tiny, it can sell as much as it wants at the ruling market price but nothing at a higher price, since buyers would go to a rival at once. The firm is therefore a **price taker**. The demand curve it faces is a horizontal line at the market price, and, as shown in Chapter 5, **AR = MR = P**. The market price itself is set by the interaction of the total demand and total supply of the industry (Chapter 2); the firm merely accepts it.

Short-run equilibrium

A perfectly competitive firm maximises profit by producing where **MC = MR = P**, with MC rising. The size of the profit depends on how the price compares with average total cost (ATC) at that output.

- If P > ATC, the firm earns **supernormal profit**, equal to (P − ATC) × Q.
- If P = ATC, it earns **normal profit** only.
- If AVC < P < ATC, it makes a loss but continues in the short run, because it covers variable cost and part of fixed cost.
- If P < AVC, it shuts down.

WORKED EXAMPLE 6.1

A perfectly competitive firm has total fixed cost of ₦40 thousand a week. Its marginal costs for the first six units of output are ₦20, ₦30, ₦40, ₦50, ₦60 and ₦80 thousand. The market price is ₦55 thousand per unit. Find the profit-maximising output and the profit.

Solution: MR = P = 55. The firm compares MC with 55 for each unit. Units 1 to 4 have MC of 20, 30, 40 and 50, all below 55, so each adds to profit. The fifth unit has MC = 60, which exceeds 55, so it would reduce profit. Output = 4 units. Total variable cost = 20 + 30 + 40 + 50 = 140. Total cost = 40 + 140 = 180. Total revenue = 55 × 4 = 220. Profit = 220 − 180 = ₦40 thousand. Average total cost = 180/4 = 45, which is below the price of 55, so the firm earns supernormal profit of (55 − 45) × 4 = ₦40 thousand, agreeing with the first calculation.

Long-run equilibrium

Supernormal profit is a signal. New firms enter the industry, the industry supply curve shifts to the right, the market price falls, and profit is competed away. If instead firms are making losses, some leave, supply falls and price rises. Entry and exit stop only when every firm earns normal profit, that is, when the price equals the lowest point of the long-run average cost curve. In long-run equilibrium therefore:

P = MR = MC = minimum ATC.

Key point. In perfect competition, long-run equilibrium has two efficiency properties. *Productive efficiency*: output is produced at the lowest possible average cost (P = minimum ATC). *Allocative efficiency*: price equals marginal cost (P = MC), so the value consumers place on the last unit exactly equals the cost of resources used to make it.

6.3 Pure monopoly

Meaning and sources of monopoly power

A **monopoly** exists when a single firm is the sole producer of a good that has no close substitutes. The firm is the industry, so the market demand curve is the firm's own demand curve. In practice a firm is often called a monopolist if it controls a very large share of the market (for example, a quarter or more by some legal definitions), even when it is not the only seller. Monopoly power arises from **barriers to entry**:

- **Legal barriers.** Patents, copyrights and exclusive licences grant sole rights for a period. In Nigeria, the generation, transmission and distribution of electricity, as well as the supply of piped water in each state, have long been operated as legal monopolies or licensed territories.
- **Control of essential raw materials.** A firm that owns the only source of a mineral or key input can exclude others.

- **Economies of scale.** Where average cost falls over a very large range of output, one big firm can supply the whole market more cheaply than several small ones. Such a case is a **natural monopoly**. Rail lines, water pipes and the national electricity grid are examples, since duplicating the network would waste resources.
- **Government ownership.** The state may keep an industry as a public monopoly for strategic or social reasons.
- **Predatory behaviour and heavy advertising** by established firms, which deter newcomers.
- **Geographical isolation or transport costs**, which give a local seller a monopoly in a small town.

Revenue and equilibrium

Since the monopolist faces the downward-sloping market demand curve, it must lower price to sell more, and MR lies below AR (Chapter 5). The monopolist maximises profit where **MC = MR**, with MC cutting MR from below, and then charges the highest price consumers will pay for that output, which is read from the demand (AR) curve. Because P > MR = MC, the price exceeds marginal cost. The monopolist is a **price maker**: it can choose the price or the quantity, but not both, since the demand curve links them.

A monopolist earns supernormal profit if AR exceeds ATC at the chosen output, and, because entry is blocked, the profit can persist in the long run. A monopolist is not guaranteed a profit: if demand is too weak to

cover ATC at any output, it too will make a loss and eventually stop production.

Another important point: a monopolist never chooses to operate on the inelastic part of its demand curve. Where demand is inelastic, MR is negative, so MC = MR cannot be satisfied (MC cannot be negative). The monopolist therefore produces where demand is elastic (PED > 1).

WORKED EXAMPLE 6.2

A monopolist faces the demand curve P = 100 − 2Q, where P is in naira and Q in units. Marginal cost is constant at ₦20, and total fixed cost is ₦200. Find (a) the profit-maximising output and price, (b) the profit, and (c) the output and price if the market were perfectly competitive, together with the loss of welfare caused by the monopoly.

Solution: (a) TR = P × Q = 100Q − 2Q². MR is the rate of change of TR, which for this straight-line demand curve is MR = 100 − 4Q (same intercept, twice the slope). Set MR = MC: 100 − 4Q = 20, so 4Q = 80 and Q = 20. Price from the demand curve: P = 100 − 2(20) = 100 − 40 = ₦60.

(b) TR = 60 × 20 = ₦1,200. TC = 200 + 20 × 20 = 200 + 400 = ₦600. Profit = 1,200 − 600 = ₦600. Check: profit per unit above MC is 60 − 20 = 40, times 20 units = 800, less fixed cost 200 = 600.

(c) In perfect competition, price equals MC: 100 − 2Q = 20, so Q = 40 and P = ₦20. The monopolist therefore produces half the competitive output (20 instead of 40) at three times the price (₦60 instead of ₦20). The **deadweight loss** is the triangle between the demand curve and MC curve, from Q = 20 to Q = 40: ½ × (40 − 20) × (60 − 20) = ½ × 20 × 40 = ₦400. This is the value of the mutually beneficial trades that do not take place.

Price discrimination

A monopolist sometimes sells the same product at different prices to different buyers, for reasons not related to differences in cost. This is **price discrimination**. Examples are cheaper cinema tickets for students, lower night-time call rates, different airfares for the same flight, and the pricing of electricity to households and to factories. For it to work, three conditions are needed:

1. The seller must have **market power** (be a price maker).
2. The market must be **separable** into groups with different elasticities of demand (by age, time, income or place).
3. There must be **no resale** (no re-selling from the cheap market to the dear market).

The rule is to equate MC with MR separately in each market. The group with the less elastic demand is charged the higher price, because that group is less sensitive to price.

WORKED EXAMPLE 6.3

A monopolist sells a product in two separated markets at a constant marginal cost of ₦10. Demand in market A is P = 50 − Q, and in market B it is P = 34 − 0.5Q. Find the output, price and profit in each market (ignore fixed costs), and say which group has the more elastic demand.

Solution: Market A: MR = 50 − 2Q. Setting MR = MC: 50 − 2Q = 10, so Q = 20, and P = 50 − 20 = ₦30. Profit = (30 − 10) × 20 = ₦400.

Market B: MR = 34 − Q. Setting MR = MC: 34 − Q = 10, so Q = 24, and P = 34 − 0.5(24) = 34 − 12 = ₦22. Profit = (22 − 10) × 24 = ₦288.

Total profit = 400 + 288 = ₦688. The price is higher in market A (₦30) than in market B (₦22), so market A has the less elastic demand. We can confirm this. In market A, Q = 50 − P, so a change of ₦1 in price changes quantity by 1 unit, and PED = 1 × 30/20 = 1.5. In market B, Q = 68 − 2P, so a ₦1 change in price changes quantity by 2 units, and PED = 2 × 22/24 ≈ 1.83. Market B is more elastic, so it is charged the lower price.

Advantages and disadvantages of monopoly

Monopoly is usually blamed for higher price, lower output, and less efficiency than competition, as Worked example 6.2 shows. Without rivals, the monopolist may also have little incentive to cut costs or

improve quality (so-called **X-inefficiency**). Income is transferred from consumers to the monopolist, and resources are tied up in defending the monopoly position.

Monopoly also has defenders. A natural monopoly can produce at lower average cost than many small firms through economies of scale. Supernormal profit may finance research and development, and patents reward invention. A monopolist may use price discrimination to supply poor consumers who would otherwise be excluded, and a state monopoly can provide essential services to remote areas where private firms would not go.

Controlling monopoly

Governments use several tools: **price controls** (setting a maximum price, as electricity regulators do through tariff orders); **taxation** of excess profit; **nationalisation**; **breaking up** large firms or preventing mergers that would create dominant firms; **encouraging competition** through licensing of new entrants (as in telecommunications, where deregulation in the early 2000s brought several mobile operators into a market once dominated by a single state-owned provider); and **regulatory agencies** that monitor prices and quality. In Nigeria, bodies such as the Nigerian Communications Commission, the Nigerian Electricity Regulatory Commission and the Federal Competition and Consumer Protection Commission carry out such duties.

6.4 Monopolistic competition

Many real markets combine a large number of sellers, which is characteristic of competition, with differentiated products, which gives each seller some monopoly power. This is **monopolistic competition**. Examples include hairdressing salons, restaurants and "buka" food sellers, boutiques, private schools, pharmacies, and brands of soap, biscuits and bottled water.

Characteristics

- **Many sellers**, each with a small share, so that one firm's actions do not noticeably affect the others and there is no collusion.
- **Product differentiation** through brand names, quality, packaging, location, service and advertising. Buyers regard the goods as close but not perfect substitutes.
- **Freedom of entry and exit** in the long run, though a new firm must usually build a reputation.
- **Non-price competition**: firms spend on advertising, design and customer service to shift demand towards their own product.

Equilibrium

Because the product is differentiated, each firm faces a downward-sloping demand curve, but one that is fairly **elastic**, since there are many close substitutes. In the short run the firm produces where MC = MR, and may earn supernormal profit or a loss.

In the long run, supernormal profit attracts new firms, whose products take away some of the existing firm's customers. The demand curve facing each firm shifts to the left and becomes more elastic until it just touches the ATC curve. At that point, **AR = ATC** at the output where MC = MR, and each firm earns only normal profit. The firm is then in long-run equilibrium: there is no tendency for firms to enter or leave.

Two features of this equilibrium deserve attention. First, price exceeds marginal cost (P > MC), so there is some allocative inefficiency. Second, the firm produces at a point on the falling part of its ATC curve, to the left of the minimum. It could lower average cost by producing more, but it cannot sell more at that price. The gap between this output and the minimum-cost output is called **excess capacity**. The "cost" of product variety is therefore a slightly higher average cost, and consumers may regard this as worth paying for the choice they have.

WORKED EXAMPLE 6.4

A monopolistically competitive bakery in Kano sells 500 loaves a day at ₦400 each in the long run. Its average total cost at that output is ₦400. Another bakery in the same town sells 300 loaves at ₦450, with average total cost ₦380. Compare their profits and say what will happen in the long run.

Solution: First bakery: AR = ATC = 400, so profit per loaf is zero and total profit is 0 × 500 = ₦0. It earns only normal profit and is in long-run equilibrium. Second bakery: profit per loaf = 450 − 380 = ₦70; total profit = 70 × 300 = ₦21,000, which is supernormal. This will attract new bakeries (or encourage rivals to imitate its product). The demand for the second bakery's loaves will fall and become more elastic; price and profit per loaf will fall until its AR equals its ATC and its supernormal profit disappears.

6.5 Oligopoly

Characteristics

Oligopoly is a market dominated by a few large firms. Its main features are:

- **A few sellers**, each large enough that its actions affect the others.
- **Interdependence.** Each firm must consider how rivals will react to a change in its price or output. This is the defining feature of

oligopoly.

- **Barriers to entry**: large capital requirements, economies of scale, brand loyalty, control of distribution networks.
- Products may be **homogeneous** (cement, cooking gas, steel) or **differentiated** (mobile networks, cars, soft drinks, banking services).
- **Heavy non-price competition**: advertising, sponsorship, free gifts, loyalty rewards and after-sales service.

Nigerian examples include mobile telecommunications, cement manufacturing, brewing and soft drinks, flour milling and air travel. The degree of concentration is measured by the **concentration ratio**, the percentage of total market sales held by the largest few firms, such as the largest four or five.

WORKED EXAMPLE 6.5

In a market the five largest firms have shares of 30%, 25%, 15%, 10% and 5%, and 40 small firms share the rest equally. Find the three-firm and five-firm concentration ratios and the share of each small firm.

Solution: Three-firm ratio = 30 + 25 + 15 = 70%. Five-firm ratio = 70 + 10 + 5 = 85%. The remaining share is 100 − 85 = 15%, shared by 40 firms, so each has 15/40 = 0.375%. A five-firm ratio of 85% shows that the market is highly concentrated, an oligopoly.

Price behaviour: the kinked demand curve

Prices in many oligopolies stay unchanged for long periods even when costs change. The **kinked demand curve** model explains this. Suppose a firm is charging a price P*. It reasons as follows.

- If it **raises** its price, rivals will *not* follow, to win its customers. The firm loses many customers, so demand above P* is **elastic**.
- If it **lowers** its price, rivals will *match* the cut to avoid losing customers. The firm gains few customers, so demand below P* is **inelastic**.

The demand curve therefore has a **kink** at P*, being flatter above it and steeper below it. Since the two segments have different slopes, the marginal revenue curve has a **vertical gap** (a discontinuity) directly below the kink. As long as the marginal cost curve passes through this gap, the profit-maximising output and price stay the same. The firm has little reason to change its price, and the price is said to be **rigid** or "sticky".

WORKED EXAMPLE 6.6

A firm in a kinked-demand oligopoly sells 10 units at the ruling price of ₦100. At 10 units, the marginal revenue from the upper (elastic) section of the demand curve is ₦70, and from the lower (inelastic) section it is ₦30. Initially MC at 10 units is ₦40. (a) If MC rises to ₦65, what happens to price and output? (b) If MC rises to ₦75, what happens?

Solution: The MR curve has a gap from ₦30 to ₦70 at an output of 10 units. (a) MC of ₦65 lies inside the gap (30 < 65 < 70). Up to 10 units, MR (70) is above MC (65), so the 10th unit is worth producing. Beyond 10 units, MR (30) is below MC (65), so the 11th is not. Output remains 10 and price stays at ₦100 despite the 62.5% rise in marginal cost (from 40 to 65). (b) MC of ₦75 is above the gap, since 75 > 70. Now the 10th unit adds 70 to revenue but 75 to cost, so it should not be produced. Output falls below 10 units and the firm moves up the elastic part of its demand curve, raising price.

Collusion and price leadership

Because competition between a few firms can be damaging (a price war may reduce everyone's profit), oligopolists are tempted to cooperate.

- A **cartel** is a formal agreement among firms to fix prices or restrict output and share the market, so that the group acts like a monopolist. The best-known international example is the **Organisation of the**

Petroleum Exporting Countries (OPEC), which sets production targets for its members; Nigeria joined in 1971. A cartel faces the problem that each member is tempted to cheat by selling more than its quota, and cartels are illegal between private firms in many countries.

- **Tacit collusion** occurs without any written agreement. In **price leadership**, one firm, usually the largest or the one with the lowest cost, announces a price change and the others follow.
- **Non-price competition** replaces price competition where firms fear price wars: they advertise, sponsor events, improve packaging and offer rewards.

Evaluation of oligopoly

Oligopoly may bring benefits: economies of scale, funds for research and development, and stable prices. But it may also bring high prices, restricted output, waste on advertising, and barriers against newcomers. Regulation is therefore directed against collusion and abuse of dominance.

6.6 Comparing the market structures

The structures can be compared in terms of their results for consumers and for efficiency.

Item	Perfect competition	Monopolistic competition	Oligopoly	Monopoly
Price relative to MC	P = MC	P > MC (slightly)	P > MC	P > MC (greatly)
Long-run output	At minimum ATC	Below minimum ATC (excess capacity)	Often below	Restricted
Product choice	None	Wide	Moderate	None
Efficiency	Productive and allocative	Neither fully	Neither fully, but may have scale gains	Neither, unless natural monopoly
Advertising	None	Important	Very important	Little, except to build image

Exam tip. A frequent question asks which feature separates two structures. For perfect competition versus monopolistic competition, the answer is *product differentiation* (identical versus differentiated goods). For monopolistic competition versus oligopoly, it is the *number of firms* and *interdependence* (many independent firms versus a few interdependent ones). For monopoly versus the others, it is *barriers to entry* and a single seller.

6.7 Long-run adjustment: a competitive example

WORKED EXAMPLE 6.7

In a perfectly competitive industry the market price is ₦50 and a typical firm's lowest long-run average cost is ₦40. The typical firm produces 200 units a week. (a) Find each firm's weekly profit. (b) What will happen in the long run, and to what level will the price tend?

Solution: (a) Profit per unit = 50 − 40 = ₦10. Weekly profit = 10 × 200 = ₦2,000, which is supernormal. (b) Supernormal profit attracts new firms, since entry is free. Industry supply shifts to the right and price falls. Entry continues until supernormal profit is eliminated, at a price equal to minimum long-run average cost, ₦40. At this price profit per unit is zero (only normal profit remains), and no further entry takes place.

Exercise 6.1

1. State four conditions that must be present for a market to be perfectly competitive.
2. A perfectly competitive firm faces a price of ₦90. Its marginal costs for units 1 to 6 are ₦30, ₦45, ₦60, ₦80, ₦105 and ₦140, and its total fixed cost is ₦100. Find the profit-maximising output and the profit.

3. A monopolist faces the demand curve P = 120 − 3Q and has a constant marginal cost of ₦30. Find the profit-maximising output and price, and the profit before deducting any fixed costs.
4. Using Question 3, find the output that a perfectly competitive industry would produce with the same demand and cost conditions, and calculate the deadweight loss from monopoly.
5. The four largest firms in a market have shares of 35%, 20%, 15% and 10%. Calculate the four-firm concentration ratio and name the likely market structure.
6. Explain why a firm in monopolistic competition has excess capacity in long-run equilibrium.

Answers to Exercise 6.1

1. Any four of: very many buyers and sellers; homogeneous product; freedom of entry and exit; perfect knowledge; perfect factor mobility; no government intervention.
2. MC < 90 for units 1 to 4 (the 4th costs 80), while the 5th costs 105 > 90, so output = 4. TR = 90 × 4 = 360. TVC = 30 + 45 + 60 + 80 = 215; TC = 100 + 215 = 315. Profit = 360 − 315 = ₦45.
3. MR = 120 − 6Q. Set MR = MC: 120 − 6Q = 30, so Q = 15. P = 120 − 3(15) = 120 − 45 = ₦75. Profit before fixed cost = (75 − 30) × 15 = 45 × 15 = ₦675.
4. P = MC: 120 − 3Q = 30, so Q = 30 and P = ₦30. Deadweight loss = ½ × (30 − 15) × (75 − 30) = ½ × 15 × 45 = ₦337.50.

5. 35 + 20 + 15 + 10 = 80%. A four-firm ratio of 80% indicates a highly concentrated market: oligopoly.
6. Each firm faces a downward-sloping demand curve that just touches its ATC curve on the falling section, so it produces less than the minimum-ATC output. It could lower its average cost by producing more but could not sell the extra output at that price.

Exercise 6.2

1. State the three conditions needed for successful price discrimination.
2. A monopolist sells 8 units at ₦50 each. To sell a ninth unit it must lower the price on all units to ₦46. Calculate the marginal revenue of the ninth unit.
3. A monopolist produces 100 units, selling each at ₦70, while the average total cost at that output is ₦55. Calculate the total profit.
4. Explain why a monopolist never produces on the inelastic section of its demand curve.
5. Describe how the kinked demand curve explains price rigidity in oligopoly.

Answers to Exercise 6.2

1. The seller has market power; markets can be separated into groups with different elasticities of demand; there is no resale between the markets.
2. TR(8) = 50 × 8 = 400. TR(9) = 46 × 9 = 414. MR = 414 − 400 = ₦14.
3. Profit per unit = 70 − 55 = ₦15; total profit = 15 × 100 = ₦1,500.
4. On the inelastic section MR is negative, but MC is always positive, so MC = MR cannot hold there. Cutting output would raise revenue and lower cost.
5. Rivals are assumed to ignore a price rise but match a price cut. Demand is therefore elastic above the going price and inelastic below it, giving a kink and a gap in MR. Changes in MC within the gap leave the profit-maximising price and output unchanged.

Chapter summary

- A market structure is defined by the number of firms, the nature of the product, barriers to entry and the degree of control over price.
- Perfect competition: many firms, identical goods, free entry, perfect knowledge. The firm is a price taker with AR = MR = P, and in long-run equilibrium P = MC = minimum ATC, giving productive and allocative efficiency.

- Monopoly: one seller, barriers to entry, price maker. It maximises profit where MC = MR, charges a price above MC, and produces less than a competitive industry, causing a deadweight loss. Natural monopolies arise from economies of scale.
- Price discrimination requires market power, separable markets and no resale. The less elastic market pays the higher price.
- Monopolistic competition: many firms with differentiated products, free entry. Long-run equilibrium gives normal profit (AR = ATC) with excess capacity.
- Oligopoly: few interdependent firms with barriers to entry. Kinked demand explains price rigidity; firms may collude through cartels (such as OPEC) or price leadership, and compete mainly by non-price means.
- Concentration ratios measure the share of the market held by the largest firms.
- Governments control market power through price regulation, taxation, competition policy, licensing and, in some cases, public ownership.

Chapter 7: National Income Accounting and Determination

Every year the National Bureau of Statistics reports how much the Nigerian economy has produced, whether the figure has grown, and which sectors have driven the change. They come from the branch of economics called **national income accounting**, which measures the total value of goods and services produced in an economy over a period, usually one year.

This chapter explains the circular flow of income, defines the main national income concepts (GDP, GNP, NDP and NNP), shows the three methods of measurement, distinguishes nominal from real values, and discusses the uses and limitations of the figures. In the last part we turn from measurement to determination: what decides the level of national income, and how a change in spending is multiplied through the economy.

7.1 The circular flow of income

An economy can be pictured as a flow of money and products between two groups: **households** and **firms**. Households own the factors of production (land, labour, capital and enterprise) and sell their services to firms. In return they receive rent, wages and salaries, interest and profit. This is the **flow of income**. Households then spend their income on the goods and services that firms produce, and this spending becomes the

revenue of firms. This is the **flow of expenditure**. The money therefore circulates: income becomes spending, spending becomes revenue, and revenue is paid out again as income.

Because every naira spent by one person is a naira received by another, three quantities must be equal in value when measured for the whole economy: the value of output, the incomes earned in producing that output, and the expenditure on that output. This identity is the foundation of all three methods of measuring national income.

Injections and leakages

The simple two-sector flow is complicated by additions to and subtractions from the flow.

- **Leakages (withdrawals)** are portions of income that are taken out of the circular flow and not passed on directly as spending on domestic output. They are *savings* (S), *taxation* (T) and *imports* (M).
- **Injections** are additions to the flow that do not come from households' spending out of current income. They are *investment* (I), *government spending* (G) and *exports* (X).

Savings placed in banks are lent to firms for investment; taxes are returned as government spending; and money paid for imports returns to Nigeria when foreigners buy exports. The economy is in equilibrium when total leakages equal total injections: $S + T + M = I + G + X$.

Key point. A three-sector or four-sector circular flow adds government and the foreign sector to households and firms. Equilibrium requires that leakages (S + T + M) equal injections (I + G + X).

7.2 National income concepts

Gross Domestic Product (GDP)

Gross Domestic Product is the total market value of all final goods and services produced *within the geographical boundaries of a country* in one year, regardless of who owns the resources that produced them. A bottling plant in Ogun State owned by a foreign company contributes to Nigeria's GDP, because the production happens inside Nigeria.

Gross National Product (GNP)

Gross National Product is the total value of final goods and services produced by the *citizens* of a country, wherever in the world they work or invest. It is obtained from GDP by adding **net factor income from abroad** (NFI):

GNP = GDP + net factor income from abroad

Net factor income from abroad is the income that Nigerians receive from abroad (profits on their foreign investments, wages of Nigerians working overseas) less the income paid to foreigners from Nigeria

(profits sent home by foreign companies, salaries of expatriates). If more income flows out than flows in, NFI is negative and GNP is smaller than GDP. Nigeria, where many foreign oil companies repatriate profits, commonly has a negative net factor income, so GNP is usually below GDP.

Net national product and net domestic product

Machinery, vehicles and buildings wear out during production. The loss in value of capital through wear and tear and obsolescence is called **depreciation** (or capital consumption). Subtracting it gives the "net" measures:

- Net Domestic Product (NDP) = GDP − depreciation
- Net National Product (NNP) = GNP − depreciation

NNP is also called **national income** in the strict sense, because it measures what the nation has actually added to its wealth after replacing the capital used up.

Market prices and factor cost

Goods are sold at **market prices**, which include indirect taxes (such as value added tax and excise duties) and may be lowered by subsidies. The income actually received by factors of production is the price less indirect taxes plus subsidies. Hence:

Value at factor cost = value at market prices − indirect taxes + subsidies

For example, NNP at factor cost = NNP at market prices − (indirect taxes − subsidies).

WORKED EXAMPLE 7.1

The following data (₦ billion) relate to an economy in one year: consumption 4,500; investment 1,200; government spending 1,500; exports 900; imports 1,100; net factor income from abroad −300; depreciation 500; indirect taxes 700; subsidies 100. Calculate (a) GDP at market prices, (b) GNP, (c) NNP at market prices, (d) NNP at factor cost.

Solution: (a) GDP = C + I + G + (X − M) = 4,500 + 1,200 + 1,500 + (900 − 1,100) = 7,200 − 200 = ₦7,000 billion. (b) GNP = GDP + NFI = 7,000 + (−300) = ₦6,700 billion. (c) NNP at market prices = GNP − depreciation = 6,700 − 500 = ₦6,200 billion. (d) Indirect taxes less subsidies = 700 − 100 = 600. NNP at factor cost = 6,200 − 600 = ₦5,600 billion.

7.3 Methods of measuring national income

Since output, income and expenditure are equal in value, national income can be measured at any of the three points of the circular flow. In principle the three methods give the same total; in practice small differences arise from data errors and are corrected by a "statistical discrepancy".

The output (product) method

This method adds up the value of the final output of every sector: agriculture, industry (mining, manufacturing, construction, utilities) and services (trade, transport, communication, finance, education and others). Only **final goods**, those bought by their ultimate users, are counted. **Intermediate goods**, which are used up in producing other goods (such as flour used by a bakery), are not counted separately; otherwise the same output would be counted several times, a mistake called **double counting**. The correct procedure is either to count only the value of final goods, or to add up the **value added** at each stage of production:

Value added = value of output − value of intermediate inputs bought from other firms

WORKED EXAMPLE 7.2

A cotton farmer in Katsina sells cotton to a ginnery for ₦20,000. The ginnery sells lint to a spinner for ₦35,000. The spinner sells yarn to a weaver for ₦55,000. The weaver sells cloth to a shop for ₦80,000, and the shop sells the cloth to consumers for ₦100,000. Calculate the value added at each stage and the contribution to GDP.

Solution: Farmer: 20,000 − 0 = ₦20,000. Ginnery: 35,000 − 20,000 = ₦15,000. Spinner: 55,000 − 35,000 = ₦20,000. Weaver: 80,000 − 55,000 = ₦25,000. Shop: 100,000 − 80,000 = ₦20,000. Total value added = 20,000 + 15,000 + 20,000 + 25,000 + 20,000 = ₦100,000. This equals the final price paid by consumers. Adding all the sales (20,000 + 35,000 + 55,000 + 80,000 + 100,000 = ₦290,000) would triple-count the cotton, and the sum would be wrong.

The income method

This method adds the incomes earned by the factors of production in producing output:

- wages and salaries (compensation of employees);
- rent;
- interest;
- profits of companies and the incomes of the self-employed (such as traders and farmers).

The total is national income at factor cost. To reach GDP at market prices one adds indirect taxes less subsidies, adds depreciation, and subtracts net factor income from abroad. Only incomes *earned* in return for productive services are included. **Transfer payments**, such as pensions, unemployment benefits, scholarships, and gifts from relatives, are excluded because no goods or services are produced in exchange for them.

The expenditure method

This method adds up all spending on final goods and services produced in the country:

$$\text{GDP} = C + I + G + (X - M)$$

where C is household consumption expenditure, I is gross investment (spending by firms on capital goods and additions to stocks, together with new residential construction), G is government spending on goods and services, X is exports and M is imports. Imports are subtracted because they are included in C, I and G but were not produced in the country. **Government transfer payments are again excluded**, as are purchases of second-hand goods (which were counted when first produced) and purchases of shares and bonds (which only transfer ownership of financial assets).

Key point. To avoid errors, remember three exclusions from national income: (1) intermediate goods, to avoid double counting; (2) transfer payments, because there is no output in exchange; (3) second-hand sales and financial transactions, because they involve no new production.

7.4 Nominal and real national income

National income measured at the prices of the year in which the output was produced is **nominal** (or money, or current-price) national income. Because prices rise in most years, nominal income may increase even when the quantity of output is unchanged. To measure genuine growth, economists adjust for price changes and express national income at the prices of a fixed **base year**. This gives **real** national income (constant-price income).

Real GDP = (nominal GDP ÷ price index) × 100

The price index used for the whole of GDP is called the **GDP deflator**:

GDP deflator = (nominal GDP ÷ real GDP) × 100

The base-year index is 100. A deflator of 120 means that the general price level is 20% higher than in the base year.

WORKED EXAMPLE 7.3

Nominal GDP of an economy was ₦50,000 million in Year 1 (the base year, price index 100) and ₦60,000 million in Year 2, when the price index was 120. Calculate the percentage growth of nominal GDP and of real GDP between the two years.

Solution: Nominal growth = (60,000 − 50,000) ÷ 50,000 × 100 = 20%. Real GDP in Year 2 = (60,000 ÷ 120) × 100 = ₦50,000 million. Real growth = (50,000 − 50,000) ÷ 50,000 × 100 = 0%. The entire rise in nominal GDP came from higher prices; the quantity of output did not grow at all.

Economic growth

Economic growth is a sustained increase in the real national output of an economy. The growth rate is measured as the percentage change in real GDP from one year to the next:

Growth rate = [(real GDP this year − real GDP last year) ÷ real GDP last year] × 100

Per capita income

Per capita income is national income divided by the population. It is the average income per person and is used to compare living standards between countries and over time.

Per capita income = national income ÷ population

WORKED EXAMPLE 7.4

An economy has a GNP of ₦72,000 billion and a population of 200 million. Calculate the per capita income. If real GNP grows by 4% while the population grows by 2.5%, approximately how fast does real per capita income grow?

Solution: GNP = $72{,}000 \times 10^9 = 7.2 \times 10^{13}$. Population = 2×10^8. Per capita income = $7.2 \times 10^{13} \div 2 \times 10^8 = 3.6 \times 10^5$ = ₦360,000. Real per capita growth ≈ 4% − 2.5% = 1.5% (more exactly $1.04 \div 1.025 = 1.0146$, or about 1.46%). Growth in per capita terms is lower than growth of total output whenever population is growing.

7.5 Uses and limitations of national income statistics

Uses

- **Measuring economic growth.** Changes in real GDP show whether the economy is expanding or contracting.
- **Comparing living standards.** Per capita income allows comparison between countries and between periods, although with caution.
- **Planning and policy.** Governments use the figures to prepare budgets, plan development and judge whether policies are working.

Limitations

- **The informal and subsistence sectors.** Much economic activity in Nigeria takes place in the informal sector (roadside traders, artisans, small farmers) and is unrecorded. Subsistence farmers consume much of what they grow, which is hard to value. National income is therefore understated and figures are only estimates.
- **Non-marketed and illegal activity.** Housework, child care and smuggling are not recorded.
- **Inflation.** Without adjustment, nominal figures overstate growth.
- **Distribution.** An average hides inequality. A high per capita income may coexist with widespread poverty if a few people receive most of the income.
- **Quality of life.** GDP does not measure leisure, health, security, education, the quality of the environment, or the depletion of resources such as oil and forests.
- **Data problems.** Weak statistical systems, differences in base years and exchange-rate movements make international comparison difficult. Statistical offices sometimes *rebase* the series, updating the base year and the weights of sectors. When Nigeria rebased its GDP in 2014 using 2010 prices, measured GDP rose sharply, not because the country had suddenly produced more but because newer sectors such as telecommunications and the film industry were now properly counted.

Economists therefore supplement national income with indicators such as the **Human Development Index**, which combines life expectancy, education and income.

Exercise 7.1

1. An economy has consumption ₦3,000 billion, investment ₦800 billion, government spending ₦1,000 billion, exports ₦600 billion and imports ₦700 billion. Calculate GDP.
2. If net factor income from abroad is −₦200 billion and depreciation is ₦300 billion, use the answer to question 1 to find GNP and NNP.
3. A farmer sells wheat to a miller for ₦10,000. The miller sells flour to a baker for ₦18,000. The baker sells bread to consumers for ₦30,000. Calculate the value added at each stage and the total contribution to GDP.
4. Nominal GDP is ₦8,400 billion and the GDP deflator is 120. Calculate real GDP.
5. A country has a GDP of ₦54 trillion and a population of 180 million. Calculate GDP per capita.
6. Explain why pensions paid by the government are not included in national income, while the salaries of civil servants are.

Answers to Exercise 7.1

1. GDP = 3,000 + 800 + 1,000 + (600 − 700) = 4,800 − 100 = ₦4,700 billion.
2. GNP = 4,700 − 200 = ₦4,500 billion. NNP = 4,500 − 300 = ₦4,200 billion.
3. Farmer: ₦10,000. Miller: 18,000 − 10,000 = ₦8,000. Baker: 30,000 − 18,000 = ₦12,000. Total = 10,000 + 8,000 + 12,000 = ₦30,000, the price of the bread.
4. Real GDP = 8,400 ÷ 120 × 100 = 70 × 100 = ₦7,000 billion.
5. $54 \times 10^{12} \div 1.8 \times 10^{8} = 30 \times 10^{4} =$ ₦300,000.
6. Pensions are transfer payments: nothing is produced in exchange for them, so including them would count the same income twice (once as tax or earlier salary, and again as pension). Civil servants' salaries are payments for services (administration, teaching, policing) that form part of government output.

7.6 Determination of national income

So far we have measured national income; now we ask what determines its level in the short run. The explanation associated with John Maynard Keynes concentrates on **aggregate demand**, the total planned spending on the country's output. When output capacity is not fully used, the level of income depends on the level of spending.

Consumption and saving

Households divide their disposable income (Y) between consumption (C) and saving (S): Y = C + S. Consumption rises as income rises, but by less than the rise in income. A simple **consumption function** is

C = a + bY

where a is **autonomous consumption**, the spending that would occur even at zero income (met from past savings or borrowing), and b is the proportion of each extra naira of income that is spent.

- **Average propensity to consume** (APC) = C ÷ Y. **Average propensity to save** (APS) = S ÷ Y. APC + APS = 1.
- **Marginal propensity to consume** (MPC) = ΔC ÷ ΔY, the fraction of an additional naira of income that is spent. **Marginal propensity to save** (MPS) = ΔS ÷ ΔY. MPC + MPS = 1.

In the function C = a + bY, the slope b is the MPC.

WORKED EXAMPLE 7.5

When a household's income rises from ₦800,000 to ₦1,000,000 a year, its consumption rises from ₦700,000 to ₦850,000. Calculate (a) the MPC, (b) the MPS, (c) the APC at an income of ₦1,000,000.

Solution: $\Delta Y = 1{,}000{,}000 - 800{,}000 = 200{,}000$. $\Delta C = 850{,}000 - 700{,}000 = 150{,}000$. (a) MPC = 150,000 ÷ 200,000 = 0.75. (b) MPS = 1 − 0.75 = 0.25. (Check: $\Delta S = 200{,}000 - 150{,}000 = 50{,}000$, and 50,000 ÷ 200,000 = 0.25.) (c) APC = 850,000 ÷ 1,000,000 = 0.85.

Equilibrium national income

In a simple two-sector economy without government or foreign trade, aggregate demand is consumption plus investment: AD = C + I. The economy is in **equilibrium** when planned expenditure equals output (income):

$$Y = C + I$$

Equivalently, since Y = C + S, equilibrium requires that planned saving equals planned investment: S = I. If planned investment exceeds saving, stocks fall and firms raise output, so income rises; if saving exceeds investment, unsold stocks build up and income falls.

For C = a + bY and a given level of investment I, substitution gives Y = a + bY + I, so Y(1 − b) = a + I, and

Equilibrium Y = (a + I) ÷ (1 − b) = (a + I) ÷ MPS

The multiplier

An increase in an injection (such as investment) raises income by a larger amount. Suppose a company spends ₦100 million building a factory. The builders and suppliers receive ₦100 million as income. If their MPC is 0.8, they spend ₦80 million on goods, which becomes income for others, who spend ₦64 million, and so on. The total increase in income is the sum of the series 100 + 80 + 64 + ... This is the **multiplier effect**. The multiplier (k) is the ratio of the change in income to the change in the injection that caused it:

k = ΔY ÷ ΔI = 1 ÷ (1 − MPC) = 1 ÷ MPS

For the example, k = 1 ÷ 0.2 = 5, so total income rises by ₦500 million.

In an open economy with government, the multiplier is smaller because there are more leakages. Where all leakages are taken into account:

k = 1 ÷ (MPS + MPT + MPM)

where MPT is the marginal propensity to tax and MPM the marginal propensity to import. A country that imports much of what it consumes has a low multiplier.

WORKED EXAMPLE 7.6

In an economy, C = 200 + 0.75Y (in ₦ billion), and planned investment is ₦100 billion. Find (a) the equilibrium national income, (b) the level of consumption and saving at equilibrium, (c) the multiplier, and (d) the new equilibrium income if investment rises by ₦40 billion.

Solution: (a) Y = (a + I) ÷ (1 − b) = (200 + 100) ÷ (1 − 0.75) = 300 ÷ 0.25 = ₦1,200 billion. (b) C = 200 + 0.75 × 1,200 = 200 + 900 = ₦1,100 billion. S = Y − C = 1,200 − 1,100 = ₦100 billion, which equals I, confirming equilibrium. (c) k = 1 ÷ 0.25 = 4. (d) ΔY = k × ΔI = 4 × 40 = ₦160 billion, so the new Y = 1,200 + 160 = ₦1,360 billion. Check: (200 + 140) ÷ 0.25 = 340 ÷ 0.25 = 1,360.

WORKED EXAMPLE 7.7

In an open economy, the marginal propensity to save is 0.2, the marginal propensity to tax is 0.1 and the marginal propensity to import is 0.2. Calculate the multiplier. By how much will national income rise if government spending increases by ₦150 billion?

Solution: Total leakage rate = 0.2 + 0.1 + 0.2 = 0.5. k = 1 ÷ 0.5 = 2. ΔY = 2 × 150 = ₦300 billion.

Inflationary and deflationary gaps

Equilibrium income need not equal the level of output at full employment. If aggregate demand at full-employment output is less than that output, there is a **deflationary gap**, and unemployment results. If aggregate demand exceeds full-employment output, there is an **inflationary gap**: the economy cannot produce more in real terms, so prices rise. The gap is measured as the amount by which aggregate demand differs from output at full employment. Governments try to close such gaps through fiscal and monetary policy.

> **Key point.** In examination questions on equilibrium, check your answer by confirming that S = I (two sectors) or that leakages equal injections (more sectors). Remember that the multiplier is the reciprocal of the *total* leakage rate, not simply of the MPS.

Exercise 7.2

1. An economy has the consumption function C = 100 + 0.8Y and investment of ₦50 billion. Find equilibrium national income.
2. If the MPC is 0.9, calculate the multiplier. What is the total change in income when investment rises by ₦30 billion?
3. Income rises from ₦500 billion to ₦700 billion and saving rises from ₦40 billion to ₦80 billion. Calculate the MPS, MPC and the multiplier.

4. At an income of ₦1,000 billion, consumption is ₦900 billion. Calculate the APC and APS.
5. In an economy the MPS is 0.1, the MPT is 0.15 and the MPM is 0.25. Calculate the multiplier and the rise in income when government spending increases by ₦120 billion.

Answers to Exercise 7.2

1. $Y = (100 + 50) \div (1 - 0.8) = 150 \div 0.2 =$ ₦750 billion. Check: $C = 100 + 600 = 700$; $S = 750 - 700 = 50 = I$.
2. $k = 1 \div (1 - 0.9) = 1 \div 0.1 = 10$. $\Delta Y = 10 \times 30 =$ ₦300 billion.
3. $\Delta S = 40$, $\Delta Y = 200$, so $MPS = 40 \div 200 = 0.2$. $MPC = 1 - 0.2 = 0.8$. $k = 1 \div 0.2 = 5$.
4. $APC = 900 \div 1{,}000 = 0.9$. $APS = 1 - 0.9 = 0.1$ (saving is ₦100 billion).
5. Leakage rate $= 0.1 + 0.15 + 0.25 = 0.5$, so $k = 2$. $\Delta Y = 2 \times 120 =$ ₦240 billion.

Chapter summary

- The circular flow shows income and expenditure moving between households and firms; leakages (savings, taxes, imports) are matched by injections (investment, government spending, exports) in equilibrium.

- GDP measures output within a country's borders; GNP = GDP + net factor income from abroad; NNP = GNP − depreciation; factor cost = market price − indirect taxes + subsidies.
- National income is measured by the output (value-added) method, the income method and the expenditure method (GDP = C + I + G + (X − M)). Intermediate goods, transfer payments, second-hand goods and financial transactions are excluded.
- Real GDP = nominal GDP ÷ price index × 100. Growth is the percentage change in real GDP; per capita income = national income ÷ population.
- National income figures are used for planning and comparison, but are limited by the informal sector, unpaid work, inflation, income inequality, and omission of quality-of-life and environmental factors.
- C = a + bY, where b is the MPC; MPC + MPS = 1 and APC + APS = 1.
- Equilibrium occurs where Y = C + I, or S = I; with C = a + bY, Y = (a + I) ÷ (1 − b).
- The multiplier is 1 ÷ MPS in the simple model, and 1 ÷ (MPS + MPT + MPM) in an open economy with taxes.

Chapter 8: Money, Inflation and Banking

A trader in Onitsha Main Market sells goods for naira, deposits the takings in a bank, pays a supplier by transfer, and borrows from a microfinance bank to restock. At each step something called money is being used, and institutions are managing it. This chapter explains what money is and what it does, how its value is related to the price level, why inflation occurs and what harm it does, and how the institutions of the financial system, from the Central Bank of Nigeria to the stock exchange, create, control and channel money.

8.1 Barter and the development of money

Before money, people practised **barter**, the direct exchange of one good for another. Barter has serious disadvantages:

- **Double coincidence of wants.** A yam farmer who wants a cutlass must find a blacksmith who wants yams at that very moment.
- **No common measure of value.** How many tubers of yam equal one goat? Every good would have a different price in terms of every other good.
- **Indivisibility.** A cow cannot be divided to pay for a bag of salt.
- **Difficulty of storing wealth and of deferred payment.** Perishable goods lose value, and promises to pay later in goods are hard to

state.

These difficulties led societies to adopt commonly accepted objects as money. In West Africa, cowrie shells, manilla (metal bracelets), brass rods and cloth served in this way. Later came coins, paper notes and, in modern economies, bank deposits and electronic payments. Nigeria replaced the Nigerian pound with the naira and kobo on 1 January 1973.

Functions of money

1. **Medium of exchange.** Money is accepted in payment for goods and services, removing the need for a double coincidence of wants.
2. **Unit of account (measure of value).** Prices are quoted in naira, which makes comparison easy.
3. **Store of value.** Money lets purchasing power be kept from one period to the next, though its value falls when prices rise.
4. **Standard of deferred payment.** Debts, rents and wages are stated in money terms and settled later.

Characteristics of good money

Good money should be generally acceptable, durable, portable, divisible, homogeneous (each unit identical), difficult to counterfeit, and stable in value, with a supply that can be regulated.

Types of money

- **Commodity money** has value in itself (gold, cowries).

- **Fiat money** is notes and coins that are money because the government declares it so and people accept it, not because the paper has any worth. The naira is fiat money. Money that must be accepted by law in settlement of debts is **legal tender**.
- **Bank (credit) money** consists of deposits in banks that can be transferred by cheque, card or electronic transfer. In developed economies it forms most of the money supply.

> **Key point.** The **money supply** is the total stock of money in circulation. A narrow definition (M1) covers currency outside banks plus demand (current account) deposits. A broader definition (M2) adds savings and time deposits.

8.2 The value of money and the quantity theory

The value of money is what it can buy. If the general price level rises, a naira buys less, so the value of money falls. The value of money is therefore the reciprocal of the price level: the higher the price level, the lower the value of money.

The classical **quantity theory of money**, in the Fisher form, links the money supply to prices using the equation of exchange:

$$MV = PT$$

where M is the quantity of money, V is the **velocity of circulation** (the average number of times a unit of money is spent in a year), P is the average price level and T is the number of transactions (or the volume of goods and services traded). If V and T are constant, P is directly proportional to M: doubling the money supply doubles the price level. The theory warns that printing money faster than output grows causes inflation. Its weakness is that V and T are not in fact constant; velocity changes with confidence and with banking habits, and output changes in the short run.

WORKED EXAMPLE 8.1

In an economy the money supply is ₦2,000 million, each naira is spent 5 times a year on average, and 2,500 million units of goods are traded. Find the price level. What happens to it if the money supply doubles while V and T are unchanged?

Solution: P = MV ÷ T = (2,000 × 5) ÷ 2,500 = 10,000 ÷ 2,500 = ₦4 per unit. After the money supply doubles to ₦4,000 million: P = (4,000 × 5) ÷ 2,500 = 20,000 ÷ 2,500 = ₦8. The price level has doubled in the same proportion as the money supply.

The demand for money

Keynes explained why people hold money, rather than assets that earn interest, by three **liquidity preference** motives:

- **Transactions motive:** to finance everyday purchases. It rises with income.

- **Precautionary motive:** to meet unexpected needs such as illness or repairs.
- **Speculative motive:** to take advantage of future changes in the prices of bonds and other assets. It varies inversely with the rate of interest, since a high rate of interest makes holding idle cash costly.

The rate of interest is the price of borrowing money. In theory it is determined where the demand for money meets the supply of money, which the central bank controls.

8.3 Inflation and deflation

Inflation is a persistent and appreciable rise in the general level of prices, so that each naira buys less. **Deflation** is a persistent fall in the general price level. **Disinflation** is a fall in the *rate* of inflation: prices still rise, but more slowly. A one-off rise in the price of a single good is not inflation.

Measuring inflation

Inflation is measured with a **consumer price index** (CPI). Statisticians select a typical "basket" of goods and services bought by households and give each item a **weight** in proportion to its share of household spending. The index is calculated as the weighted average of the price relatives (current price ÷ base-year price × 100):

Index = Σ(weight × price index of item) ÷ Σ(weights)

The inflation rate is the percentage change in the index between two periods:

Inflation rate = [(CPI this year − CPI last year) ÷ CPI last year] × 100

WORKED EXAMPLE 8.2

A simple consumer basket has three components. Food has weight 60 and a price index of 130; clothing has weight 25 and a price index of 110; transport has weight 15 and a price index of 120 (base year = 100). Calculate the consumer price index. If the index was 105.0 a year earlier on the same basis, calculate the rate of inflation.

Solution: Index = (60 × 130 + 25 × 110 + 15 × 120) ÷ (60 + 25 + 15) = (7,800 + 2,750 + 1,800) ÷ 100 = 12,350 ÷ 100 = 123.5. Inflation = (123.5 − 105.0) ÷ 105.0 × 100 = 18.5 ÷ 105 × 100 = 17.6% (to one decimal place).

Causes of inflation

- **Demand-pull inflation:** total spending exceeds the economy's capacity to produce, so "too much money chases too few goods". It may follow an increase in the money supply, heavy government spending financed by borrowing from the central bank, or a rise in export earnings.
- **Cost-push inflation:** rising costs of production force firms to raise prices. Sources include higher wages, higher fuel prices, increased import prices following depreciation of the naira, and higher taxes.

- **Imported inflation:** a country that depends on imported food, machinery and fuel imports higher prices when world prices rise or when the exchange rate falls.
- **Structural causes:** poor roads, insecurity and inadequate storage raise the cost of moving food from farms to cities, so that supply is inelastic.

Effects of inflation

- **Fixed-income earners** (pensioners, salaried workers whose pay is slow to adjust) and **lenders** lose, since the real value of their money falls. **Borrowers** gain because they repay with money worth less.
- **Savings** lose real value if the interest rate is below the inflation rate; saving is discouraged.
- Exports become less competitive if domestic prices rise faster than those abroad, worsening the balance of payments.
- Uncertainty discourages long-term investment, and resources move into speculation (land, foreign currency) rather than production.
- Firms bear *menu costs* of changing prices, and workers' demands for higher pay may raise costs in turn.

Controlling inflation

- **Monetary policy:** raising interest rates, increasing reserve requirements, and selling securities to reduce the money supply.
- **Fiscal policy:** reducing government spending and raising taxes to lower aggregate demand.

- **Supply-side measures:** improving agricultural output, transport and power supply, and removing bottlenecks, which reduce costs and raise supply.
- **Prices and incomes policy, and import liberalisation:** controlling wage growth, or permitting more imports of scarce goods.

Deflation is not simply good news. Falling prices cause consumers to delay purchases and make debts harder to repay, which can reduce output and employment.

WORKED EXAMPLE 8.3

A savings account pays 18% a year while inflation is 12%. A second account pays 10% while inflation is 15%. Find the approximate real rate of interest in each case.

Solution: Real interest rate ≈ nominal rate − inflation rate. First account: 18 − 12 = 6% (an exact calculation gives 1.18 ÷ 1.12 − 1 = 5.4%). Second account: 10 − 15 = −5%. In the second case the saver's money loses purchasing power even though interest is paid, because prices rise faster than the balance grows.

Exercise 8.1

1. State four functions of money and explain why a barter economy cannot easily perform them.
2. In an economy M = ₦500 billion, V = 6 and T = 1,500 billion units. Use MV = PT to find the price level.

3. A basket consists of food (weight 50, price index 140), housing (weight 30, price index 120) and transport (weight 20, price index 110). Calculate the consumer price index.
4. A price index rises from 160 to 184 in one year. What is the rate of inflation?
5. Distinguish demand-pull inflation from cost-push inflation, and give one Nigerian example of each.

Answers to Exercise 8.1

1. Medium of exchange, unit of account, store of value and standard of deferred payment. Barter needs a double coincidence of wants, has no common measure of value, cannot divide indivisible goods, and cannot store value or settle debts easily, since goods are perishable and varying in value.
2. P = MV ÷ T = (500 × 6) ÷ 1,500 = 3,000 ÷ 1,500 = ₦2.
3. Index = (50 × 140 + 30 × 120 + 20 × 110) ÷ 100 = (7,000 + 3,600 + 2,200) ÷ 100 = 12,800 ÷ 100 = 128.
4. (184 − 160) ÷ 160 × 100 = 24 ÷ 160 × 100 = 15%.
5. Demand-pull: total demand exceeds supply, for example heavy government spending financed by new money. Cost-push: costs rise, for example a rise in the pump price of petrol that lifts transport fares and food prices. (Any sensible examples.)

8.4 The financial system

The financial system channels the savings of surplus units (households with money to spare) to deficit units (firms and governments that need funds). Its institutions fall into groups.

The Central Bank of Nigeria (CBN)

The central bank is the apex institution of the financial system. The CBN was established by the Central Bank of Nigeria Act of 1958 and began operations in 1959. Its main functions are:

1. **Sole issuer of legal tender**: it issues naira notes and coins.
2. **Banker and financial adviser to the government**: it keeps government accounts, makes payments, manages public debt and advises on economic policy.
3. **Banker to the commercial banks**: banks keep reserve deposits with it, and it clears inter-bank payments.
4. **Lender of last resort**: it lends to banks that are short of cash and cannot borrow elsewhere, preventing panics and bank failures.
5. **Custodian of foreign reserves** and manager of exchange-rate policy.
6. **Regulator and supervisor** of banks and other financial institutions.
7. **Conduct of monetary policy**: controlling the money supply and credit to achieve price stability and growth.

Instruments of monetary policy

- **Open market operations (OMO):** buying or selling government securities. Selling securities takes money from banks and reduces credit; buying has the opposite effect.
- **Monetary Policy Rate (MPR):** the benchmark rate at which the central bank lends to banks. A higher MPR pushes up commercial lending rates and discourages borrowing.
- **Cash reserve ratio (CRR):** the percentage of deposits that banks must hold with the CBN. A higher ratio reduces the funds available for lending.
- **Liquidity ratio:** the minimum proportion of deposits that must be held in liquid assets such as cash, treasury bills and balances with the CBN.
- **Moral suasion and directives:** persuasion, or direct instructions on lending, for example the sectors to which banks should lend.

A policy that reduces the supply of money and raises interest rates is called **tight (contractionary)** policy and is used against inflation. A policy that increases the money supply and lowers interest rates is **loose (expansionary)** policy and is used to stimulate output in a recession.

Commercial banks

Commercial banks, such as the large deposit-taking banks in every Nigerian city, accept deposits and lend. Their functions include accepting deposits (current, savings and fixed or time deposits), granting loans and overdrafts, discounting bills, transferring money, providing

foreign exchange services, and offering safe custody, advice and agency services. A bank must balance three aims: **profitability** (lending at interest), **liquidity** (being able to repay depositors on demand) and **safety** (avoiding bad debts). Its assets, ranked from most liquid and least profitable to least liquid and most profitable, are cash, balances with the central bank, money at call, treasury bills, bills discounted, investments and, finally, loans and advances.

Other financial institutions

- **Development banks** (such as the Bank of Industry and the Nigerian Export-Import Bank) give long-term, low-interest loans for industry and exports, which commercial banks may be reluctant to provide.
- **Microfinance banks** make small loans to market women, artisans and small farmers who lack collateral.
- **Mortgage banks** finance housing; **insurance companies** pool risk and invest premiums; **pension fund administrators** invest retirement savings.
- **Discount houses** and the money market institutions trade in short-term securities.

Money market and capital market

The **money market** deals in short-term funds, up to one year. Its instruments include treasury bills, commercial paper, certificates of deposit and inter-bank loans. The **capital market** deals in long-term funds, for periods longer than one year, through shares (equities), government and corporate bonds, and debentures. Shares are traded on

the stock exchange, the Nigerian Exchange. The *primary* market issues new securities, and the *secondary* market is where existing securities are bought and sold, which gives investors liquidity.

8.5 Credit creation

Banks cannot lend all the money deposited, since depositors may demand cash. They keep a fraction as a **reserve** and lend the rest. When that loan is spent, the recipient pays it into a bank, which can lend most of it again. A single initial deposit therefore supports a much larger quantity of deposits. This process is called **credit creation**.

If banks hold a fraction r of deposits as reserves, the maximum total deposits supported by an initial cash deposit D are

Maximum total deposits = D ÷ r, and credit created = D ÷ r − D

The factor 1 ÷ r is the **money multiplier**. Actual credit creation is smaller if customers hold some of their money as cash, or if banks keep extra reserves or find too few good borrowers.

WORKED EXAMPLE 8.4

A customer deposits ₦10 million cash in a bank. All banks must keep 20% of deposits as reserves. Calculate the maximum total deposits and the credit created, and show the first three stages.

Solution: r = 0.20, so the money multiplier = 1 ÷ 0.20 = 5. Maximum total deposits = 10 ÷ 0.20 = ₦50 million. Credit created = 50 − 10 = ₦40 million. Stages: the first bank keeps ₦2 million and lends ₦8 million. That is redeposited; the next bank keeps ₦1.6 million and lends ₦6.4 million; the next keeps ₦1.28 million and lends ₦5.12 million. Total loans = 8 + 6.4 + 5.12 + ... = 8 ÷ (1 − 0.8) = ₦40 million, which agrees with the formula.

WORKED EXAMPLE 8.5

Banks hold total deposits of ₦200 billion, and the CBN's required cash reserve ratio is 30%. The CBN raises the ratio to 35%. Calculate the change in required reserves and the effect on the amount banks can lend, assuming they keep only the required reserves.

Solution: Required reserves before = 0.30 × 200 = ₦60 billion; after = 0.35 × 200 = ₦70 billion. The rise is ₦10 billion. Lendable funds before = 200 − 60 = ₦140 billion; after = 200 − 70 = ₦130 billion, a fall of ₦10 billion. By reducing banks' lending capacity, the higher ratio tightens credit and is deflationary.

Key point. In examination problems on credit creation, divide the initial deposit by the reserve ratio to get the maximum total deposits, then subtract the initial deposit to get the additional credit. Check whether the question asks for total deposits or for the credit created.

Exercise 8.2

1. A bank receives a cash deposit of ₦50 million. If the reserve ratio is 10%, calculate the maximum total deposits and the credit created.
2. A bank has deposits of ₦400 million and holds ₦60 million in cash. If the required cash ratio is 12%, calculate its excess reserves.
3. Name three instruments the CBN can use to reduce the money supply and say how each works.
4. Find the approximate real rate of interest where the nominal rate is 14% and inflation 9%, and where the nominal rate is 8% and inflation 12%.
5. Distinguish the money market from the capital market, giving one instrument of each.
6. Explain what is meant by the central bank acting as "lender of last resort".

Answers to Exercise 8.2

1. Total deposits = 50 ÷ 0.10 = ₦500 million. Credit created = 500 − 50 = ₦450 million.
2. Required reserves = 0.12 × 400 = ₦48 million. Excess = 60 − 48 = ₦12 million.
3. Selling securities through open market operations (banks pay for them with reserves, so lending falls); raising the Monetary Policy Rate (borrowing from the CBN and loans to customers become dearer); raising the cash reserve ratio (less of deposits can be lent).
4. 14 − 9 = 5%; 8 − 12 = −4% (savers lose purchasing power).
5. The money market deals in short-term funds of up to a year (treasury bills); the capital market deals in long-term funds (shares or bonds traded on the stock exchange).
6. When a bank cannot borrow elsewhere to meet withdrawals, the central bank lends to it (usually at a penal rate), which maintains confidence in the banking system and stops a failure spreading.

Chapter summary

- Barter requires a double coincidence of wants and lacks a common measure; money solves these problems by serving as a medium of exchange, unit of account, store of value and standard of deferred payment.
- Good money is acceptable, durable, portable, divisible, homogeneous, scarce and stable. Modern money is fiat money

and bank deposits.

- The quantity theory MV = PT shows prices rising with the money supply when V and T are constant. Keynes's liquidity preference identifies transactions, precautionary and speculative motives for holding money.
- Inflation is a persistent rise in the general price level, measured by a weighted consumer price index. Demand-pull, cost-push, imported and structural factors cause it; it hurts fixed-income earners, lenders and savers and discourages investment.
- Inflation is controlled by monetary, fiscal and supply-side measures. Real interest rate ≈ nominal rate − inflation rate.
- The CBN issues currency, banks the government and the banks, acts as lender of last resort, holds reserves, supervises banks and conducts monetary policy through OMO, the MPR, the cash reserve ratio, the liquidity ratio and moral suasion.
- Commercial banks balance profitability, liquidity and safety. Development, microfinance, mortgage and insurance institutions serve special needs; the money market handles short-term funds and the capital market long-term funds.
- Credit creation: maximum deposits = initial deposit ÷ reserve ratio; credit created = maximum deposits − initial deposit.

Chapter 9: Public Finance and Fiscal Policy

Roads, public schools, police stations, the salaries of nurses and soldiers, and the subsidies on some fertilisers all depend on government money. Where does that money come from, how is it spent, and what happens to the economy when the government spends more than it receives? **Public finance** is the study of how governments raise revenue, spend it, and manage their budgets and debts. **Fiscal policy** is the deliberate use of government revenue and expenditure to influence the level of economic activity, prices and employment.

In this chapter we examine why governments are needed in the economy, the sources and classification of revenue with emphasis on taxation, the pattern of public spending, budgets and public debt, and the way fiscal policy is used. Nigeria is a federation of three tiers of government (federal, state and local), and this shapes how revenue is shared.

9.1 The role of government in the economy

Markets work well for many goods, but they fail in several ways, and government intervenes to correct the failure.

- **Public goods.** A public good, such as national defence, street lighting or flood control, is *non-excludable* (people who have not

paid cannot be kept from using it) and *non-rival* (one person's use does not reduce what is left for others). Private firms will not supply such goods because they cannot charge those who benefit, so government provides them from taxation.

- **Merit goods.** Goods such as education and immunisation are beneficial, but people tend to consume too little of them if left to the market. Government provides or subsidises them.
- **Externalities.** Production or consumption may impose costs (pollution from gas flaring in the Niger Delta) or confer benefits (vaccination) on people outside the transaction. Government uses taxes, regulation or subsidies to bring private decisions closer to social interest.
- **Monopoly power** (Chapter 6) and the need to protect consumers.
- **Stabilisation and equity.** Government tries to keep output growing, prices stable and employment high, and redistributes income from rich to poor through taxation and social spending.

9.2 Sources of government revenue

Public revenue comes from taxes and from non-tax sources.

Taxation

A **tax** is a compulsory payment made by individuals and firms to government, for which no direct, specific service is received in return. Taxes are divided into two main groups.

- **Direct taxes** are levied on income or wealth and are paid by the person on whom they are imposed. Examples: personal income tax (including Pay-As-You-Earn, PAYE, deducted from salaries), company income tax, capital gains tax, petroleum profit tax and tenement rates.
- **Indirect taxes** are levied on spending, on goods and services, and are collected by sellers who pass the burden to buyers in higher prices. Examples: value added tax (VAT, at 7.5% in Nigeria), excise duties on goods such as tobacco and beer, and import duties (customs tariffs).

Classification by the relation of tax to income

- A **progressive tax** takes a higher percentage of income as income rises (as in a graduated personal income tax). It reduces inequality.
- A **proportional tax** takes the same percentage at all income levels.
- A **regressive tax** takes a smaller percentage of income as income rises. Many indirect taxes are regressive, because the poor spend a larger share of their income on taxed necessities.

The **average tax rate** is total tax ÷ total income × 100. The **marginal tax rate** is the tax on an additional naira of income, and for graduated taxes it is the rate on the highest band reached.

WORKED EXAMPLE 9.1

An illustrative income tax schedule applies a rate of 5% to the first ₦400,000 of annual taxable income, 10% to the next ₦400,000, and 20% to all income above ₦800,000. Calculate the tax payable by Amina, who earns ₦1,200,000, and find her average and marginal tax rates.

Solution: First band: 5% of 400,000 = ₦20,000. Second band: 10% of 400,000 = ₦40,000. Third band: income above 800,000 is 1,200,000 − 800,000 = 400,000, taxed at 20% = ₦80,000. Total tax = 20,000 + 40,000 + 80,000 = ₦140,000. Average rate = 140,000 ÷ 1,200,000 × 100 = 11.67%. Marginal rate = 20%. Because the average rate (11.67%) rises as income rises towards 20%, the tax is progressive.

WORKED EXAMPLE 9.2

VAT is charged at 7.5%. (a) A shopkeeper sells a television for ₦40,000 before VAT. What is the VAT and the price paid by the customer? (b) A customer pays ₦53,750 for a refrigerator, VAT included. What was the price before VAT, and how much VAT was paid?

Solution: (a) VAT = 7.5% × 40,000 = ₦3,000. Customer pays 40,000 + 3,000 = ₦43,000. (b) The price including VAT is 107.5% of the price before VAT, so price before VAT = 53,750 ÷ 1.075 = ₦50,000. VAT = 53,750 − 50,000 = ₦3,750. Check: 7.5% × 50,000 = 3,750.

WORKED EXAMPLE 9.3

The following table shows the tax paid by three households. Classify the tax.

Income (₦)	Tax (₦)
100,000	5,000
200,000	8,000
400,000	12,000

Solution: Tax as a percentage of income: 5,000 ÷ 100,000 = 5%; 8,000 ÷ 200,000 = 4%; 12,000 ÷ 400,000 = 3%. The percentage taken falls as income rises, so the tax is regressive.

Canons (principles) of a good tax

Adam Smith proposed four principles that remain the standard test of a tax system.

1. **Equity (ability to pay):** people should pay according to their ability, so that the burden is fairly shared.
2. **Certainty:** the taxpayer should know how much to pay, when and how.
3. **Convenience:** payment should be made at a time and in a manner convenient to the taxpayer, as with PAYE deducted from pay.
4. **Economy:** the cost of collection should be small compared with the amount collected.

Modern writers add **productivity** (adequate yield), **flexibility** (the rate can be changed easily) and **neutrality** (minimal distortion of economic decisions).

Effects of taxation

High taxes on income may reduce work effort and saving, and high taxes on profits may reduce investment. Taxes on goods raise prices and reduce quantity demanded, the more so where demand is elastic (Chapter 3). Taxes can also be used deliberately to discourage harmful consumption (for example on cigarettes), to protect infant industries (tariffs), to redistribute income and to restrain demand in inflation. The

incidence of a tax, the question of who finally bears it, depends on the elasticities of demand and supply: where demand is inelastic, consumers bear most of the burden.

Non-tax revenue

- **Oil revenue:** royalties, rents and the profits of the state-owned oil company, which for Nigeria are the largest single source of federal revenue.
- **Fees, fines, licences and rents** from government property.
- **Surpluses of public enterprises** and earnings from the sale of government assets.
- **Grants and aid** from other governments and international organisations.
- **Borrowing** from domestic and foreign lenders, which is not strictly revenue but finances the deficit.

Revenue allocation in Nigeria

Revenue collected by the federal government (including oil revenue, VAT and import duties) is paid into the **Federation Account** and shared among the three tiers of government by a formula set by the Revenue Mobilisation Allocation and Fiscal Commission (RMAFC) and approved under the constitution. The principles used include equality of states, population, social development, land mass and terrain, and **derivation**, under which oil- and gas-producing states receive at least

13% of the revenue from the resources produced in them. The principles are intended to balance fairness against the need to reward effort in revenue generation.

Key point. Direct taxes are paid by the person on whom they fall (the burden cannot easily be shifted). Indirect taxes are collected from sellers but borne wholly or partly by consumers. In questions on progressive, proportional and regressive taxes, always compare the *percentage* of income taken, not the amount paid.

Exercise 9.1

1. Classify as direct or indirect: PAYE, VAT, import duty, company income tax, excise duty on beer.
2. A tax schedule takes 10% of the first ₦500,000 and 20% of anything above. Calculate the tax payable on an income of ₦1,100,000 and the average rate.
3. VAT at 7.5% is charged on a generator priced ₦200,000 before tax. Find the VAT and the total price.
4. A radio costs ₦32,250 including 7.5% VAT. What is the price before VAT?
5. State Adam Smith's four canons of taxation. Why is a tax on salt, which every household buys in similar quantities, likely to be regressive?

6. Define a public good and explain why private firms will not supply it.

Answers to Exercise 9.1

1. PAYE: direct. VAT: indirect. Import duty: indirect. Company income tax: direct. Excise duty: indirect.
2. First band tax = 0.10 × 500,000 = ₦50,000. Second band = 0.20 × (1,100,000 − 500,000) = 0.20 × 600,000 = ₦120,000. Total = ₦170,000. Average rate = 170,000 ÷ 1,100,000 × 100 = 15.45%.
3. VAT = 0.075 × 200,000 = ₦15,000. Total = ₦215,000.
4. 32,250 ÷ 1.075 = ₦30,000 (VAT = ₦2,250).
5. Equity, certainty, convenience and economy. A poor household and a rich one pay roughly the same tax on salt, so the tax is a much larger share of the poor household's income.
6. A public good is non-excludable and non-rival, so those who do not pay cannot be prevented from using it. Firms cannot recover their costs through sales, so government supplies it from tax revenue.

9.3 Public expenditure

Government spending is classified in two main ways.

- **Recurrent expenditure** covers the regular, day-to-day costs of running government: salaries and wages, pensions, overheads, subsidies and the interest on debts.

- **Capital expenditure** is spending on assets that last: roads, bridges, hospitals, power stations, schools and machinery. It raises the productive capacity of the economy.

A second classification divides public spending into **transfer payments** (pensions, grants and subsidies, which move income from one group to another without any output in exchange) and **purchases of goods and services**.

Causes of the growth of public expenditure

- Rising population and urbanisation increase the demand for schools, hospitals, water and roads.
- Inflation raises the naira cost of the same programmes, and salary reviews raise the wage bill.
- Greater demands for security and defence.
- Development plans and the government's commitment to industrialisation.
- Rising interest on public debt.
- Political and administrative expansion, including the creation of new states, ministries and agencies.

Effects of public expenditure

Capital spending on infrastructure raises productivity and attracts private investment. Government spending creates jobs and, through the multiplier, raises income. But excessive recurrent spending, waste and

corruption reduce the benefits, and spending financed by borrowing can raise interest rates and inflation.

9.4 The budget and public debt

A **budget** is a statement of the government's estimated revenue and planned expenditure for a financial year. In Nigeria the executive prepares the budget and presents it to the National Assembly for approval as an Appropriation Act. A budget may be:

- **balanced** when planned revenue equals planned expenditure;
- in **surplus** when revenue exceeds expenditure;
- in **deficit** when expenditure exceeds revenue.

Budget balance = revenue − expenditure. A negative result is a deficit.

Financing a deficit

- **Domestic borrowing** by selling treasury bills and bonds to banks, pension funds and the public. This may push up interest rates and **crowd out** private borrowers, who are left with less credit.
- **External borrowing** from foreign governments, the World Bank, the IMF, the African Development Bank and international bond markets. It brings in foreign exchange but adds debt-service burdens in foreign currency; if the naira depreciates, the naira cost of repayment rises.
- **Borrowing from the central bank** (printing money, "ways and means" advances). This raises the money supply directly and is

highly inflationary.

- **Drawing on savings** or selling assets (privatisation).

The accumulation of past deficits is the **public (national) debt**: the total amount owed by government to domestic and foreign lenders. In Nigeria the Debt Management Office (DMO) records and manages it. Two common measures of the burden are the **debt-to-GDP ratio** and the **debt-service ratio**, which is the share of government revenue (or export earnings) used to pay interest and principal.

Public debt is not harmful in itself: borrowing that finances productive capital projects with a return higher than the interest cost leaves the country better off. It becomes a problem when borrowing finances consumption, when the debt service takes up a large share of revenue and leaves little for schools and hospitals, or when it is mostly in foreign currency.

WORKED EXAMPLE 9.4

A government plans revenue of ₦7.5 trillion, recurrent expenditure of ₦5.2 trillion and capital expenditure of ₦3.8 trillion. GDP is ₦150 trillion. Debt service of ₦1.5 trillion is included in the recurrent figure. (a) Find the budget balance. (b) Express the deficit as a percentage of GDP. (c) Find the share of revenue taken by debt service.

Solution: (a) Total expenditure = 5.2 + 3.8 = ₦9.0 trillion. Balance = 7.5 − 9.0 = −₦1.5 trillion, a deficit of ₦1.5 trillion. (b) 1.5 ÷ 150 × 100 = 1%. (c) Debt service ratio = 1.5 ÷ 7.5 × 100 = 20% of revenue.

9.5 Fiscal policy

Fiscal policy uses taxation and public spending to influence aggregate demand, employment, prices and growth.

- **Expansionary fiscal policy** raises government spending or reduces taxes. It increases aggregate demand and is used to fight unemployment and recession (a deflationary gap).
- **Contractionary fiscal policy** cuts spending or raises taxes to reduce aggregate demand and control inflation (an inflationary gap).

Some elements of fiscal policy are **automatic stabilisers**: in a boom, tax receipts rise and welfare spending falls without any new decision, damping demand; in a slump the reverse happens. Fiscal policy is also

used to affect the structure of the economy, through tax holidays and incentives for firms in priority sectors, and subsidies to farmers. Its limitations include time lags in recognising problems, passing budgets and seeing the effects; political pressures; the weak links between spending and results; crowding out; and the difficulty of cutting spending or raising taxes in an election year. Fiscal policy works alongside the monetary policy described in Chapter 8.

Fiscal multipliers

From Chapter 7, the multiplier for a change in government spending is k = 1 ÷ (1 − MPC). A change in taxes works less strongly because part of a tax cut is saved at first rather than spent. The **tax multiplier** is −MPC ÷ (1 − MPC): negative because a rise in tax lowers income, and smaller in size than the spending multiplier by exactly 1.

WORKED EXAMPLE 9.5

The MPC is 0.75. The government is considering either (a) raising spending by ₦90 billion or (b) cutting taxes by ₦90 billion. Calculate the change in national income in each case. What happens if the government raises both spending and taxes by ₦90 billion?

Solution: Spending multiplier = 1 ÷ (1 − 0.75) = 4. (a) $\Delta Y = 4 \times 90 =$ ₦360 billion. The tax multiplier has size 0.75 ÷ 0.25 = 3. (b) A tax cut raises income by 3 × 90 = ₦270 billion. If both spending and taxes rise by ₦90 billion: $\Delta Y = +360 - 270 =$ ₦90 billion, an increase equal to the amount of the extra spending. This is called the balanced-budget multiplier, which equals 1.

WORKED EXAMPLE 9.6

Explain with figures why borrowing from the central bank is more inflationary than borrowing from the public. Suppose a government finances a ₦200 billion deficit in the two ways, and assume banks hold a 25% reserve ratio against deposits.

Solution: If the government sells bonds to the public for ₦200 billion, money simply passes from private buyers to government, and total money supply is unchanged at first: the government spends what others have given up. If instead the central bank creates ₦200 billion of new money for the government, the money supply rises by ₦200 billion at once. When it is spent and deposited, banks can lend against it; with a 25% reserve ratio, the money multiplier is $1 \div 0.25 = 4$, so deposits could in the end rise by as much as $4 \times 200 = ₦800$ billion. This large expansion of money is why such financing is inflationary.

Exercise 9.2

1. A government expects revenue of ₦7 trillion, recurrent spending of ₦5 trillion and capital spending of ₦3.5 trillion. Find the budget balance.
2. If GDP is ₦100 trillion and the deficit is ₦3 trillion, express the deficit as a percentage of GDP.
3. Debt service is ₦480 billion and revenue is ₦3,200 billion. Calculate the debt-service ratio.

4. The MPC is 0.8. By how much does income change if (a) government spending rises by ₦50 billion; (b) taxes are cut by ₦50 billion?
5. Distinguish recurrent expenditure from capital expenditure and give two examples of each.
6. Explain "crowding out" and say why borrowing from the central bank can cause inflation.

Answers to Exercise 9.2

1. Expenditure = 5 + 3.5 = ₦8.5 trillion; balance = 7 − 8.5 = −₦1.5 trillion (deficit of ₦1.5 trillion).
2. 3 ÷ 100 × 100 = 3%.
3. 480 ÷ 3,200 × 100 = 15%.
4. (a) k = 1 ÷ 0.2 = 5, so ΔY = 5 × 50 = ₦250 billion. (b) Tax multiplier = 0.8 ÷ 0.2 = 4, so ΔY = 4 × 50 = ₦200 billion.
5. Recurrent: regular running costs, for example salaries and pensions. Capital: spending on lasting assets, for example roads and hospitals.
6. Crowding out: government borrowing raises interest rates or absorbs available loanable funds so that private firms invest less. Borrowing from the central bank creates new money, which increases the money supply directly and, through bank credit creation, raises demand faster than output.

Chapter summary

- Government intervenes to provide public and merit goods, correct externalities, curb monopoly, stabilise the economy and redistribute income.
- Taxes are direct (income, company profits) or indirect (VAT, excise, import duties); they may be progressive, proportional or regressive depending on how the percentage of income taken varies with income.
- A good tax meets Smith's canons of equity, certainty, convenience and economy. Incidence depends on elasticities.
- Non-tax revenue includes oil income, fees, grants and the surpluses of public enterprises; revenue from the Federation Account is shared among the three tiers of government, with derivation for resource-producing states.
- Expenditure is recurrent or capital; it grows with population, prices, security needs and debt interest.
- Budgets may be balanced, in surplus or in deficit. Deficits are financed by domestic or external borrowing, or by money creation, which is the most inflationary. Debt problems arise when debt service takes a large share of revenue.
- Fiscal policy is expansionary (more spending, lower taxes) or contractionary. The spending multiplier is $1 \div (1 - \text{MPC})$, the tax multiplier is $-\text{MPC} \div (1 - \text{MPC})$, and the balanced-budget multiplier is 1.

Chapter 10: International Trade, Balance of Payments and Exchange Rates

The phone in your pocket may have been assembled in Asia from parts made in several countries. The petrol in a Lagos filling station may come from crude oil that Nigeria exported and then re-imported as refined product. Nigerian cocoa is drunk as chocolate in Europe, and Nigerian students watch films produced in the United States and India. No country produces everything it needs, and international trade is the exchange of goods and services across national boundaries. This chapter explains why countries trade, who gains, how governments restrict trade, how a country records its dealings with the rest of the world in the balance of payments, and how the exchange rate links the currencies of trading nations.

10.1 Why countries trade

International trade (foreign trade) is trade between residents of different countries. Countries trade because:

- **Resources are unevenly distributed.** Nigeria has crude oil, Ghana has cocoa and gold, Saudi Arabia has oil but little water, and Japan has little land but great skills. Climate, soil, minerals, labour skills and capital differ.

- **Specialisation raises output.** Countries that concentrate on what they produce at the lowest opportunity cost obtain more in total.
- **Wider markets allow economies of scale** (Chapter 4). A firm that sells to many countries can produce in larger quantities at a lower cost per unit.
- **Variety and competition.** Consumers get more choice, and domestic firms are pressed to be efficient.
- **Technology transfer** and access to inputs that cannot be produced at home.

International versus domestic trade

Trade between countries differs from trade within a country because of different currencies (and the need for foreign exchange), restrictions such as tariffs and quotas, differences in language, laws and standards, more limited mobility of labour and capital, and higher transport and insurance costs.

10.2 Absolute and comparative advantage

A country has an **absolute advantage** in a good if it can produce it using fewer resources than another country, that is, more output from the same resources. A country has a **comparative advantage** in a good if it can produce it at a lower *opportunity cost* than another country, that is, by giving up less of another good.

The theory of comparative advantage, developed by David Ricardo, states that even if one country is more efficient in producing everything (absolute advantage in all goods), both countries still gain if each specialises in the good in which it has a comparative advantage, and trades. What matters is not who is better in absolute terms, but the relative cost.

WORKED EXAMPLE 10.1

In one day, using all its resources, Nigeria can produce either 60 tonnes of rice or 30 tonnes of cloth. In the same time Ghana can produce either 20 tonnes of rice or 20 tonnes of cloth. (a) Which country has an absolute advantage in each good? (b) Calculate the opportunity cost of each good in each country. (c) Which country has the comparative advantage in each good?

Solution: (a) Nigeria produces more of both goods (60 against 20 rice; 30 against 20 cloth), so Nigeria has an absolute advantage in both. (b) Nigeria: producing 60 rice means giving up 30 cloth, so 1 tonne of rice costs 30 ÷ 60 = 0.5 tonne of cloth, and 1 tonne of cloth costs 60 ÷ 30 = 2 tonnes of rice. Ghana: 1 tonne of rice costs 20 ÷ 20 = 1 tonne of cloth, and 1 tonne of cloth costs 1 tonne of rice. (c) Rice: Nigeria's opportunity cost (0.5 cloth) is lower than Ghana's (1 cloth), so Nigeria has the comparative advantage in rice. Cloth: Ghana's opportunity cost (1 rice) is lower than Nigeria's (2 rice), so Ghana has the comparative advantage in cloth.

WORKED EXAMPLE 10.2

Continuing from Example 10.1, suppose that before trade each country splits its resources equally between the two goods. Show that specialisation and trade can increase total output.

Solution: Before trade: Nigeria produces half of each: 30 tonnes of rice and 15 tonnes of cloth. Ghana produces 10 tonnes of rice and 10 tonnes of cloth. Total = 40 tonnes of rice and 25 tonnes of cloth. After specialisation: Ghana produces only cloth: 20 tonnes. Nigeria must supply the remaining 25 − 20 = 5 tonnes of cloth, which uses 5 ÷ 30 = 1/6 of its resources, leaving 5/6 for rice: 5/6 × 60 = 50 tonnes. Total = 50 tonnes of rice and 25 tonnes of cloth. The same amount of cloth is produced as before, and rice has increased from 40 to 50 tonnes, a gain of 10 tonnes. Trade then divides the gain between the two countries.

The terms of trade between the two countries

Both countries gain only if the rate at which they exchange lies between their two opportunity costs. In the example, Nigeria will be willing to export rice only if it receives more than 0.5 tonne of cloth for each tonne of rice (its cost at home), and Ghana will import rice only if it pays less than 1 tonne of cloth (its cost of making rice at home). So any rate between 0.5 and 1 tonne of cloth per tonne of rice benefits both. The nearer the rate is to 1, the more Nigeria gains.

Assumptions and limitations

The theory assumes two countries and two goods, no transport costs, free trade, constant costs and full mobility of resources within a country. In reality transport costs and tariffs reduce the gains; costs may rise as production expands; and specialising in one or two primary exports exposes a country to price swings in world markets, as Nigeria has experienced with crude oil. Developing countries may be locked into low-value primary products while industrial countries sell manufactures.

> **Key point.** To find comparative advantage, compare opportunity costs, not total output. A country with an absolute advantage in everything still has a comparative advantage in only some goods. The country with the lower opportunity cost of a good should specialise in it.

10.3 The terms of trade

The **terms of trade** measure the rate at which a country's exports exchange for its imports. The most common measure is the **net barter (commodity) terms of trade**:

Terms of trade = (index of export prices ÷ index of import prices) × 100

An increase means that the country can buy more imports for a given volume of exports: the terms have **improved** (favourable). A fall means the terms have **deteriorated** (unfavourable). Countries that export

primary products and import manufactured goods often suffer from deteriorating terms of trade, because the prices of primary goods are unstable and tend to rise more slowly than those of manufactures. A country such as Nigeria whose exports depend on oil is affected sharply when the world oil price falls.

WORKED EXAMPLE 10.3

In a base year, a country's export and import price indices are both 100. Five years later the export price index is 110 and the import price index is 125. Calculate the terms of trade and say what has happened to them. If the export price index had instead risen to 130 and the import price index to 104, what would the terms of trade be?

Solution: First case: terms of trade = 110 ÷ 125 × 100 = 88. They have fallen by 12 points, so they have deteriorated: the country must export more to buy the same imports. Second case: 130 ÷ 104 × 100 = 125. The terms have improved by 25%.

10.4 Restrictions on trade

Free trade is trade without restrictions. **Protection** is the use of government measures to shield domestic producers from foreign competition. The main instruments are:

- **Tariffs (customs duties):** taxes on imports. They raise the price of imports and bring revenue.

- **Quotas:** limits on the quantity of a good that may be imported.
- **Embargoes and import bans:** total prohibition of trade in certain goods or with certain countries.
- **Subsidies** to domestic producers, which lower their costs so that they can compete.
- **Exchange controls and import licensing:** restricting the foreign currency available for imports, or requiring permits.
- **Non-tariff barriers:** technical standards, health regulations and slow customs procedures.

WORKED EXAMPLE 10.4

A foreign-made fan costs ₦80,000 at the port. The government imposes a 25% import tariff. (a) What is the new price before other costs? (b) If 5,000 fans are still imported, how much revenue does government earn?

Solution: (a) Tariff per fan = 25% × 80,000 = ₦20,000. New price = 80,000 + 20,000 = ₦100,000. (b) Revenue = 5,000 × 20,000 = ₦100,000,000 (₦100 million). If the higher price causes fewer fans to be imported, revenue is lower; and domestic consumers pay more.

Arguments for protection

- **Infant industries:** new industries need temporary protection until they grow large enough to compete.

- **Dumping:** foreign firms may sell goods abroad below cost or below the home price to eliminate competitors; tariffs counter this.
- **Balance of payments:** restricting imports can reduce a deficit.
- **Employment:** protection of local jobs.
- **Diversification:** reducing dependence on a few exports.
- **National security:** keeping vital industries such as food or defence under national control.
- **Revenue:** tariffs are easy to collect.

Arguments against protection

- It raises prices and restricts consumer choice.
- It shields inefficient firms from competition, and infant industries may never "grow up".
- Trading partners may retaliate, harming exporters.
- It reduces the gains from specialisation, lowering world output.
- It can encourage smuggling, as with goods crossing Nigeria's land borders.

Trading blocs and international organisations

Countries often reduce barriers within groups. In an **economic community** or **free trade area** members remove tariffs among themselves; in a **customs union** they also adopt a common external tariff; a **common market** adds free movement of labour and capital. Examples: the Economic Community of West African States (ECOWAS), the African Continental Free Trade Area (AfCFTA) and the

European Union. The **World Trade Organization (WTO)** sets rules for trade and settles disputes; the **International Monetary Fund (IMF)** promotes monetary cooperation and lends to countries with balance of payments difficulties; the **World Bank** provides finance for development; the **African Development Bank** funds African projects; and the **Organisation of the Petroleum Exporting Countries (OPEC)**, of which Nigeria is a member, coordinates oil policy.

Exercise 10.1

1. In one day Country A can produce 40 tonnes of maize or 20 tonnes of yams; Country B can produce 30 tonnes of maize or 30 tonnes of yams. Calculate the opportunity cost of each good in each country and state which country has the comparative advantage in each.
2. Between what limits must the exchange rate of maize for yams lie in Question 1 for both countries to gain from trade?
3. The export price index is 130 and the import price index is 104. Calculate the terms of trade.
4. An imported generator costs ₦60,000 before tariff. A tariff of 30% is imposed. Find the new price, and the revenue earned if 2,000 are imported.
5. Distinguish between absolute and comparative advantage.
6. Give four arguments for protection and two against it.

Answers to Exercise 10.1

1. A: 1 tonne of maize costs 20 ÷ 40 = 0.5 tonne of yams; 1 tonne of yams costs 2 tonnes of maize. B: 1 tonne of maize costs 1 tonne of yams; 1 tonne of yams costs 1 tonne of maize. A has the comparative advantage in maize (0.5 < 1), B in yams (1 < 2).
2. Between 0.5 and 1 tonne of yams per tonne of maize (A will not accept less than 0.5; B will not pay more than 1).
3. 130 ÷ 104 × 100 = 125, an improvement.
4. Tariff = 0.30 × 60,000 = ₦18,000. New price = ₦78,000. Revenue = 2,000 × 18,000 = ₦36,000,000.
5. Absolute advantage: producing a good with fewer resources (more output per unit of input) than another country. Comparative advantage: producing it at a lower opportunity cost.
6. For: infant industries, dumping, balance of payments, employment, diversification, security. Against: higher prices and less choice, inefficiency, retaliation, lower world output, smuggling. (Any four and two.)

10.5 The balance of payments

The **balance of payments (BOP)** is a systematic record of all economic transactions between the residents of a country and the rest of the world over a period, usually a year. It is divided into accounts.

The current account

The current account records trade in goods and services, income from abroad and transfers:

- **Trade in goods** (visible trade): exports and imports of goods such as crude oil, cocoa, cars and machinery. The balance is the **balance of trade**: exports of goods minus imports of goods.
- **Trade in services** (invisible trade): shipping, insurance, tourism, banking and telecommunications.
- **Primary income:** profits, interest, dividends and wages paid to or received from abroad.
- **Secondary income (transfers):** gifts, aid and the remittances that Nigerians abroad send to relatives at home.

The capital and financial accounts

The **capital account** records capital transfers and the acquisition of non-produced assets. The **financial account** records changes in ownership of financial assets: foreign direct investment (such as a foreign company building a factory in Nigeria), portfolio investment (buying shares and bonds) and loans, together with changes in official **reserves** of foreign currency. Money flowing into the country is entered as a credit and money flowing out as a debit.

Key point. The balance of trade covers only visible goods. The current account balance also includes services, income and transfers. Because the books must balance, a current account deficit must be financed by a surplus on the capital and financial accounts (borrowing or selling assets) or by drawing down reserves.

WORKED EXAMPLE 10.5

A country records the following (₦ billion): exports of goods 12,000; imports of goods 14,500; net services −1,200; net primary income −900; net transfers (remittances) +3,000; net capital and financial inflows +1,300. Calculate (a) the balance of trade, (b) the current account balance, (c) the overall balance and how it is financed.

Solution: (a) Balance of trade = 12,000 − 14,500 = −₦2,500 billion (a deficit). (b) Current account = −2,500 − 1,200 − 900 + 3,000 = −₦1,600 billion. (c) Overall balance = −1,600 + 1,300 = −₦300 billion. The deficit of ₦300 billion must be financed by reducing foreign exchange reserves or by further borrowing.

Disequilibrium and its causes

A **deficit** arises when payments abroad exceed receipts. Its causes include heavy dependence on imports, a fall in export prices (for Nigeria, the price of oil), high domestic inflation making exports

uncompetitive, an overvalued currency, capital flight and poor export performance. A **surplus** arises when receipts exceed payments and can build up reserves.

Correcting a deficit

- **Expenditure-reducing policies:** contractionary fiscal and monetary policy reduce incomes and imports, at the cost of lower growth and jobs.
- **Expenditure-switching policies:** measures to divert demand from imports to home output, including tariffs and quotas, devaluation or depreciation, and import-substitution.
- **Export promotion:** export incentives and diversification into agriculture, solid minerals and manufacturing.
- **Exchange controls** and external borrowing.
- **Attracting foreign investment**, including remittances.

10.6 Exchange rates

The **exchange rate** is the price of one currency in terms of another, for example ₦1,000 = US$1. Importers need foreign currency to pay for imports, and exporters receive foreign currency that they sell for naira. The exchange rate is therefore determined, in a market, by the demand for and supply of the currency.

- In a **fixed exchange rate system** the central bank keeps the rate at an official level by buying or selling foreign exchange from its

reserves.

- In a **floating (flexible) system** the rate is set by market forces.
- In a **managed float** the rate floats but the central bank intervenes to smooth sharp swings. Nigeria's exchange-rate policy has moved between these regimes over time.

A fall in the value of a currency under a floating system is a **depreciation**; a deliberate, official lowering of a fixed rate is a **devaluation**. The reverse movements are **appreciation** and **revaluation**.

Factors affecting the exchange rate

- The demand for imports and the demand for exports.
- Relative inflation: a country with higher inflation than its partners tends to see its currency depreciate.
- Interest rates: higher rates attract foreign funds and raise the demand for the currency.
- Capital flows, including foreign investment and remittances, and confidence in the economy.
- Government and central bank intervention and the level of reserves.
- Speculation.

Effects of depreciation

A depreciation makes imports dearer in naira and exports cheaper in foreign currency. If demand for exports and imports is elastic enough (the **Marshall–Lerner condition**: the sum of the price elasticities of

demand for exports and imports exceeds 1), the volume changes are large enough to improve the trade balance. If demand is inelastic, as it is for essential imports, the import bill simply rises and the deficit may worsen. Depreciation also adds to cost-push inflation and raises the naira cost of repaying foreign debt.

WORKED EXAMPLE 10.6

The exchange rate moves from ₦1,000 = $1 to ₦1,250 = $1. (a) What happens to the naira price of a machine costing $200? (b) What is the dollar price of a Nigerian export priced at ₦500,000 before and after? (c) By what percentage has the naira lost value in dollar terms?

Solution: (a) Before: 200 × 1,000 = ₦200,000. After: 200 × 1,250 = ₦250,000, a rise of 50,000 ÷ 200,000 × 100 = 25%. (b) Before: 500,000 ÷ 1,000 = $500. After: 500,000 ÷ 1,250 = $400. The export is 20% cheaper for foreign buyers. (c) One naira was worth $0.001 and is now worth 1 ÷ 1,250 = $0.0008, a fall of 0.0002 ÷ 0.001 × 100 = 20%. The naira price of imports rises by 25% while the naira's dollar value falls by 20%: the two percentages differ because they use different bases.

Exercise 10.2

1. A country's exports of goods are ₦8,000 billion and imports of goods are ₦9,500 billion. Net services are −₦500 billion and

net transfers are +₦1,200 billion. Calculate the balance of trade and the current account balance (ignore primary income).

2. The exchange rate is ₦1,500 = $1. An imported machine costs $40. What is its naira price? What would it cost if the rate became ₦1,600 = $1?
3. A Nigerian export is priced at ₦800,000. Find its dollar price at ₦1,000 = $1 and at ₦1,600 = $1.
4. Distinguish between the balance of trade and the balance of payments.
5. Explain how a depreciation of the naira might improve the balance of payments, and why it might not.
6. List four possible causes of a balance of payments deficit and two ways of correcting it.

Answers to Exercise 10.2

1. Balance of trade = 8,000 − 9,500 = −₦1,500 billion. Current account = −1,500 − 500 + 1,200 = −₦800 billion.
2. 40 × 1,500 = ₦60,000; 40 × 1,600 = ₦64,000, a rise of 4,000 ÷ 60,000 = 6.7%.
3. 800,000 ÷ 1,000 = $800; 800,000 ÷ 1,600 = $500.
4. The balance of trade covers only visible exports and imports of goods. The balance of payments records all transactions with the rest of the world: goods, services, income, transfers and capital and financial flows.

5. Exports become cheaper abroad and imports dearer at home, so export volumes may rise and import volumes fall. If the sum of the elasticities of demand is below 1 (essential imports, inelastic exports), the higher import cost can outweigh the volume change, and the deficit can worsen.
6. Causes: import dependence, falling oil prices, domestic inflation, an overvalued currency, capital flight. Corrections: export promotion, import substitution, contractionary policy, tariffs or devaluation. (Any four and two.)

Chapter summary

- Countries trade because resources are unevenly distributed, to gain from specialisation, economies of scale, variety and competition.
- Absolute advantage means producing with fewer resources; comparative advantage means a lower opportunity cost. Both countries gain from specialising according to comparative advantage and trading at a rate between their opportunity costs.
- The terms of trade = (export price index ÷ import price index) × 100; a rise is an improvement, and a fall a deterioration.
- Protection uses tariffs, quotas, embargoes, subsidies and exchange controls. It is defended on grounds such as infant industries and dumping, but it raises prices, invites retaliation and reduces efficiency.

- Trading blocs (ECOWAS, AfCFTA) and bodies such as the WTO, IMF, World Bank and OPEC shape the trading environment.
- The balance of payments has a current account (goods, services, income, transfers) and capital and financial accounts. The balance of trade is only the goods part of the current account.
- Deficits are corrected by expenditure-reducing and expenditure-switching policies and export promotion.
- The exchange rate is the price of one currency in terms of another. Depreciation makes imports dearer and exports cheaper; its effect on the trade balance depends on elasticities (Marshall–Lerner condition).